

模拟电子技术基础

Fundamentals of Analog Circuits

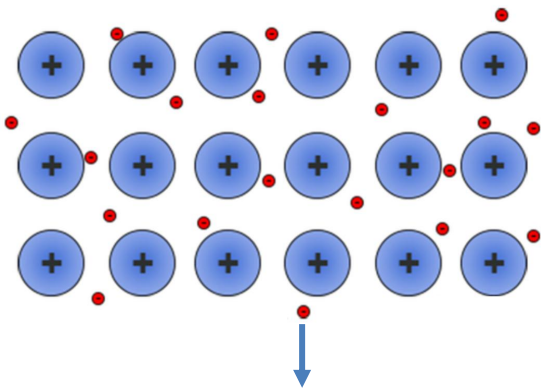
1. 半导体材料 Semiconductor

陈晓龙
Chen, Xiaolong

1.0 What is semiconductor? 什么是半导体

Metal 金属

High conductivity

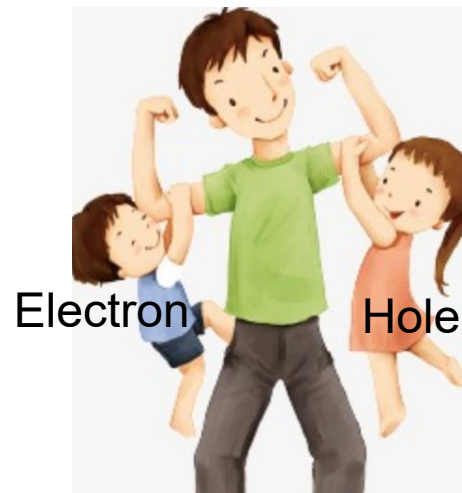
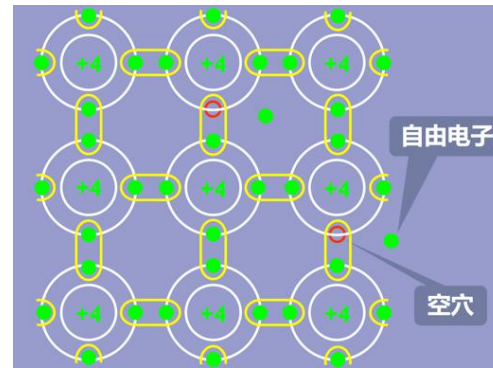


Free electron 自由电子



Semiconductor

Medium conductivity



Insulator 绝缘体

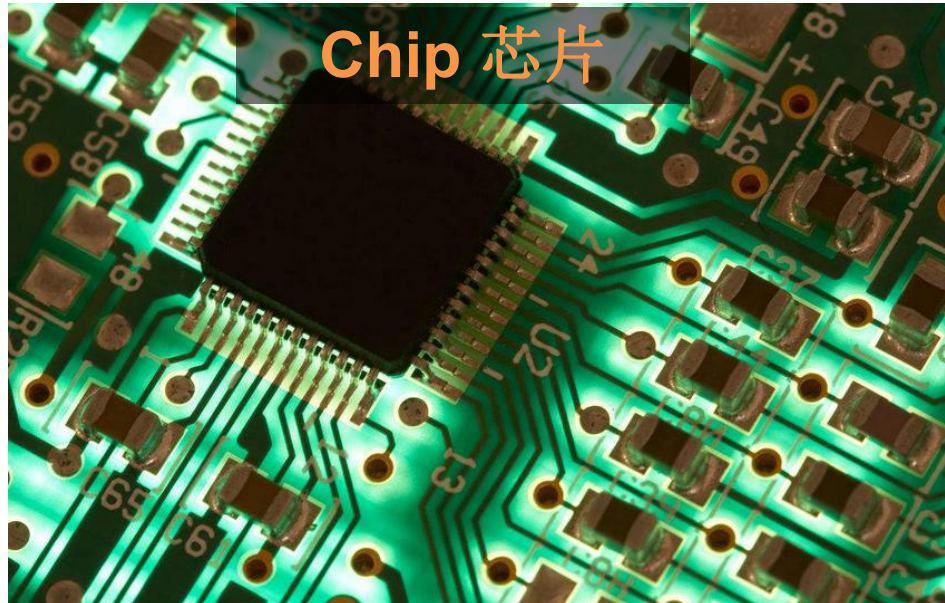
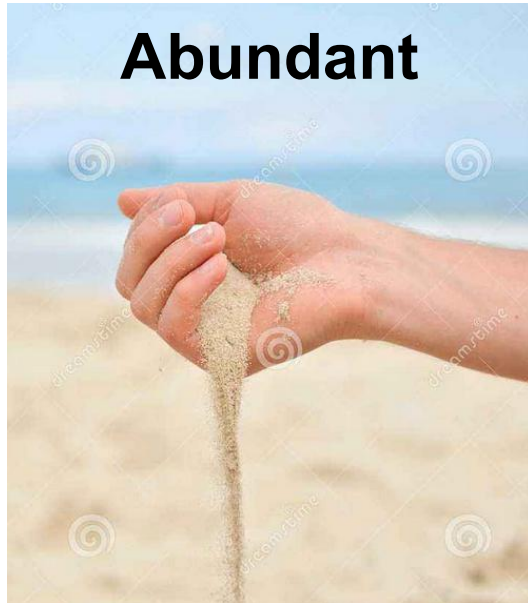
Extremely low conductivity
电导率极低

Tightly-bonded electron
束缚电子



◆ Most investigated and popular semiconductor is

Silicon 硅 and Germanium 锗

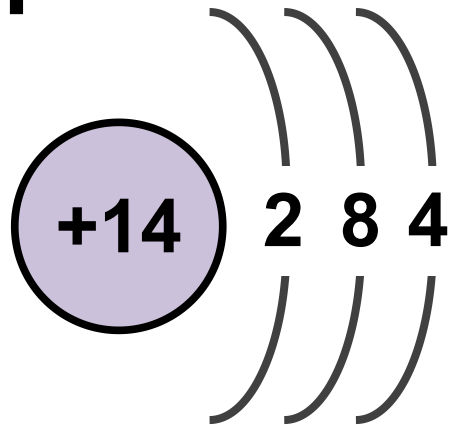


Silicon 硅 : King of the electronic materials

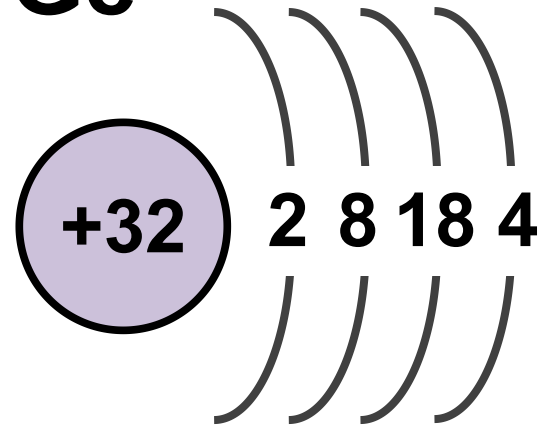


Structure of Silicon and Germanium atoms

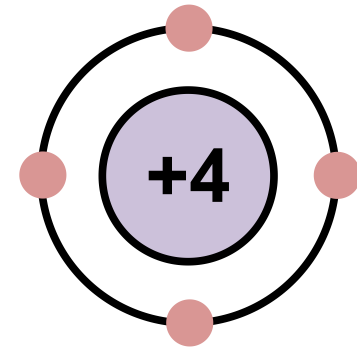
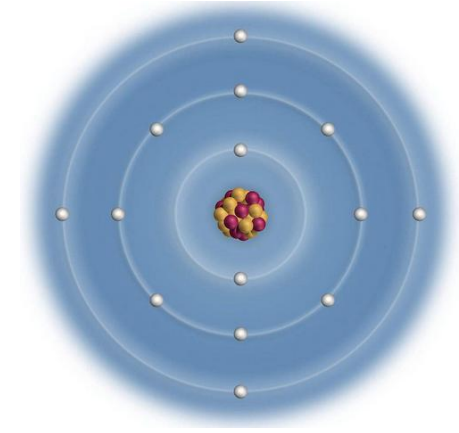
Si



Ge



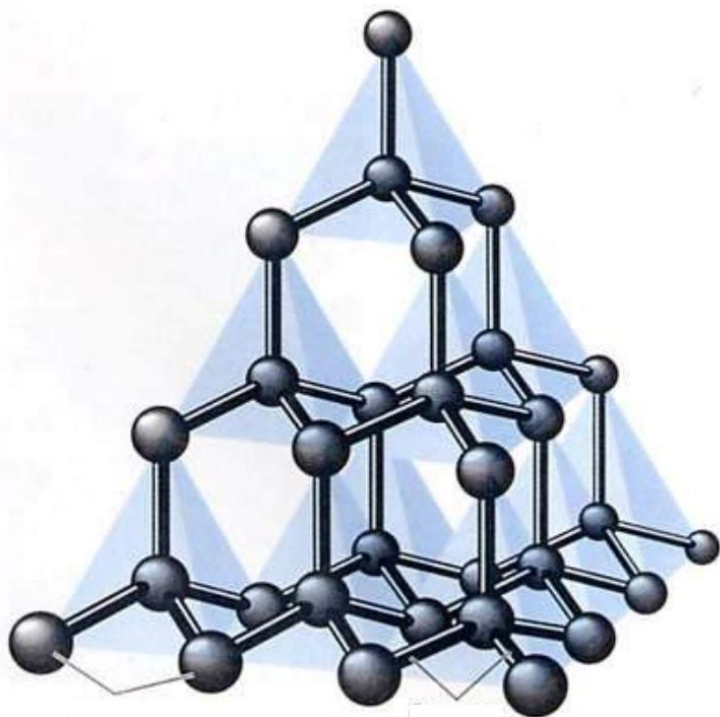
Only outmost shell is not fully filled
只有最外层电子没有填满



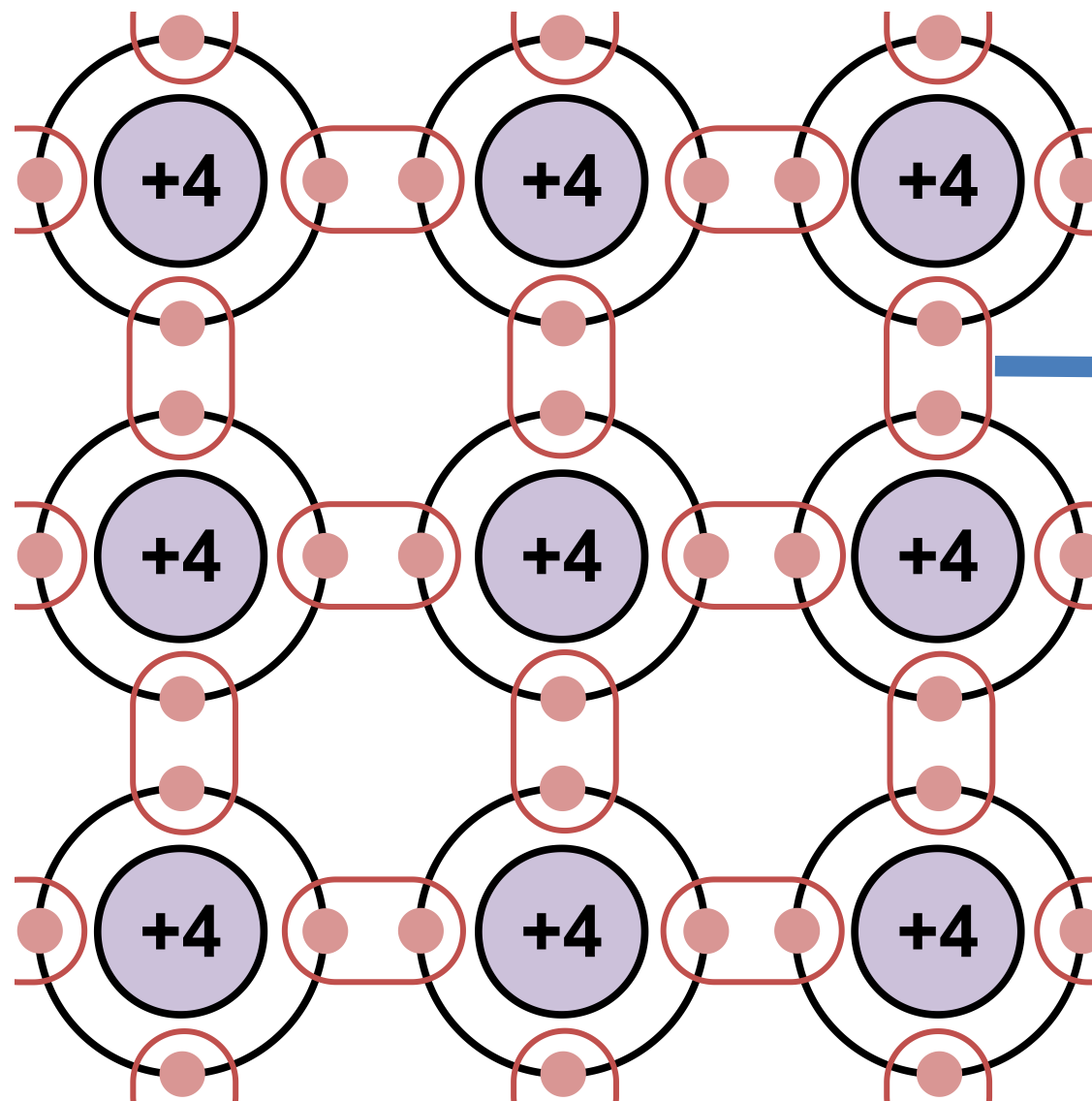
Simplified atomic structure
简化原子模型

1.1 Intrinsic semiconductor: Si/Ge 本征硅/锗半导体

3D structure 三维结构

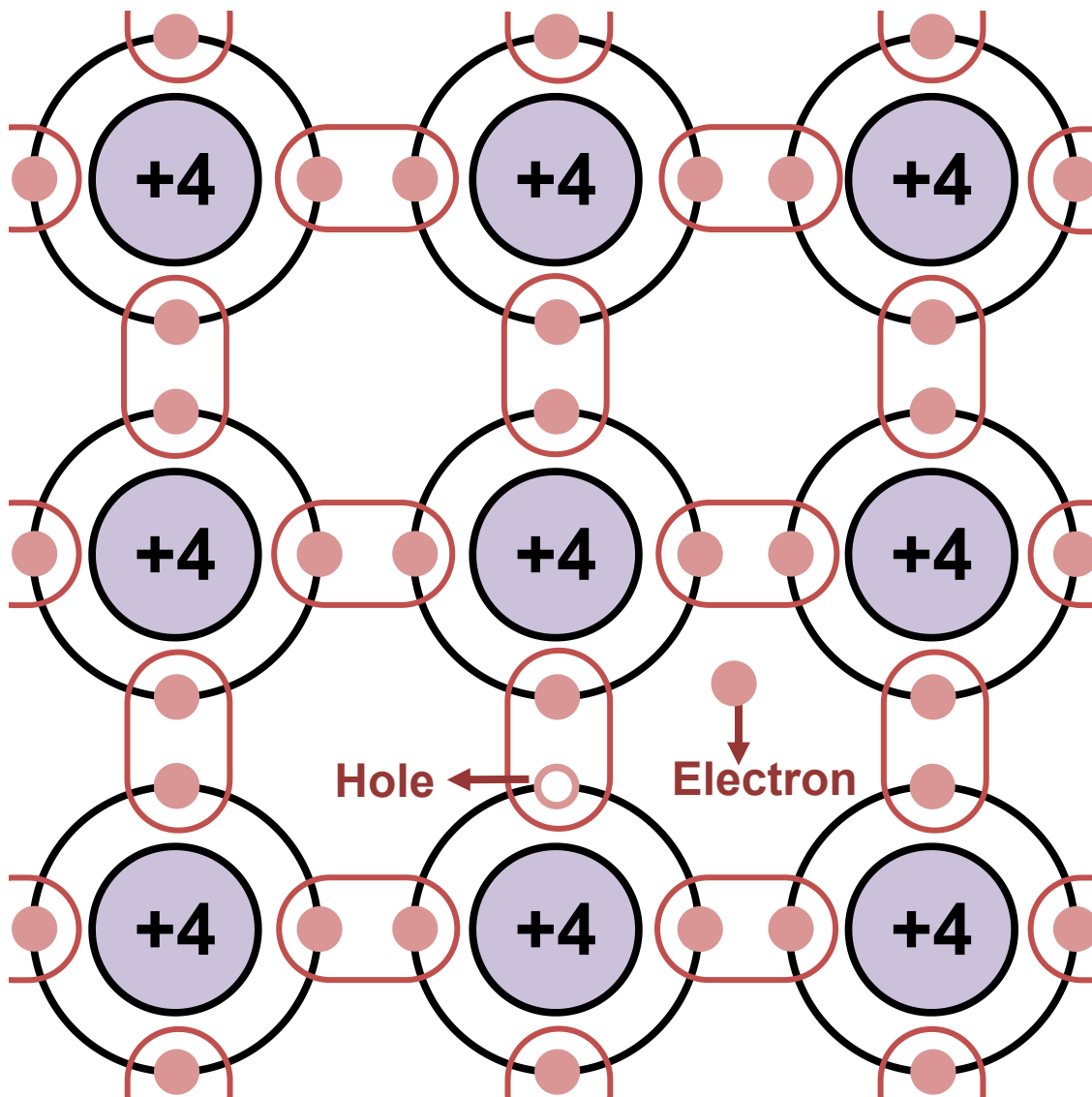


共价键
Covalent bond



2D equivalent
structure
二维等效结构

共价键
Covalent bond



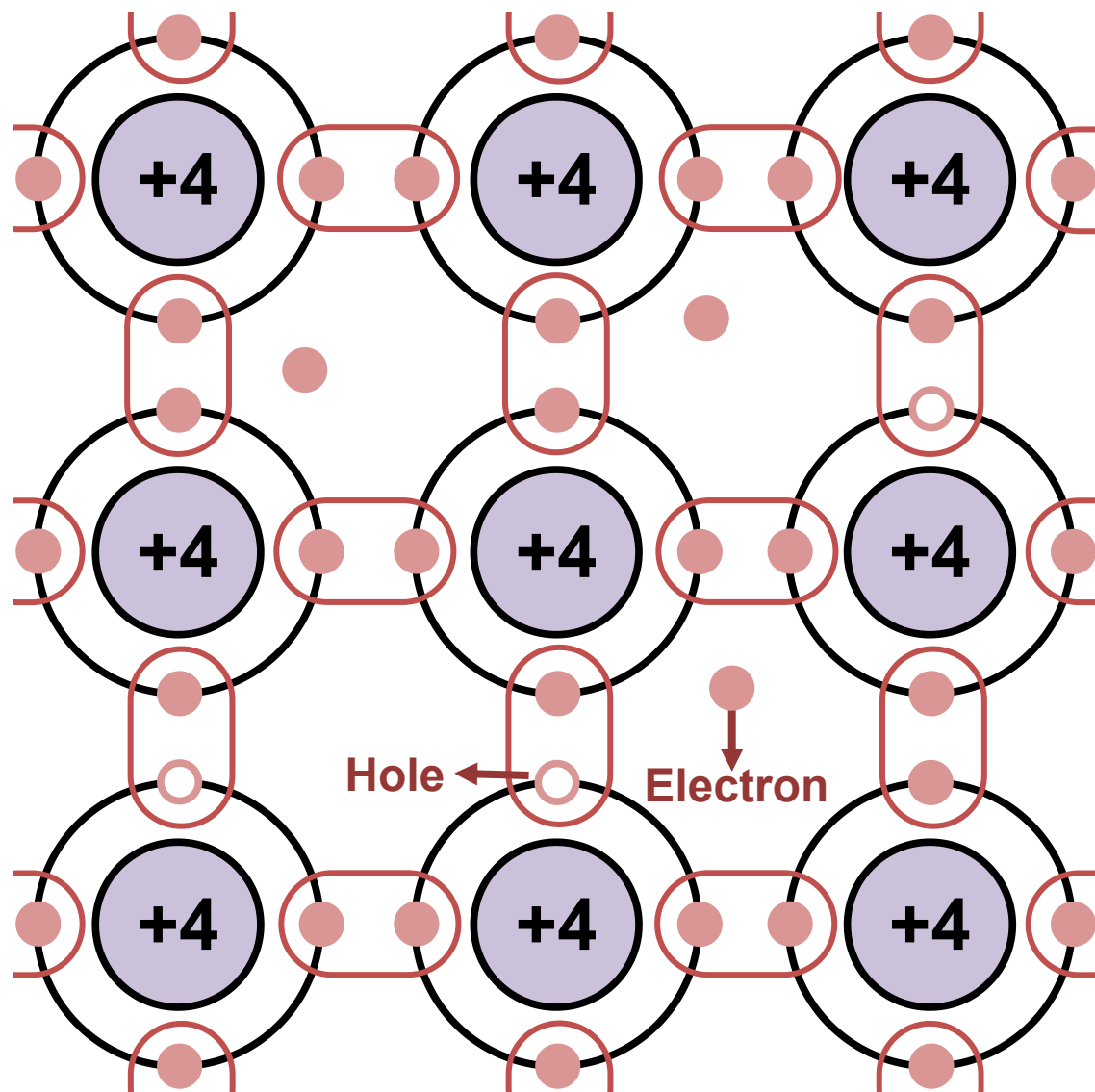
Thermal / light activation
热 / 光激发



Bonded electrons are excited
电子吸收能量脱离束缚，被激发



Electron and hole pair excitation
电子空穴对激发



Higher temperature

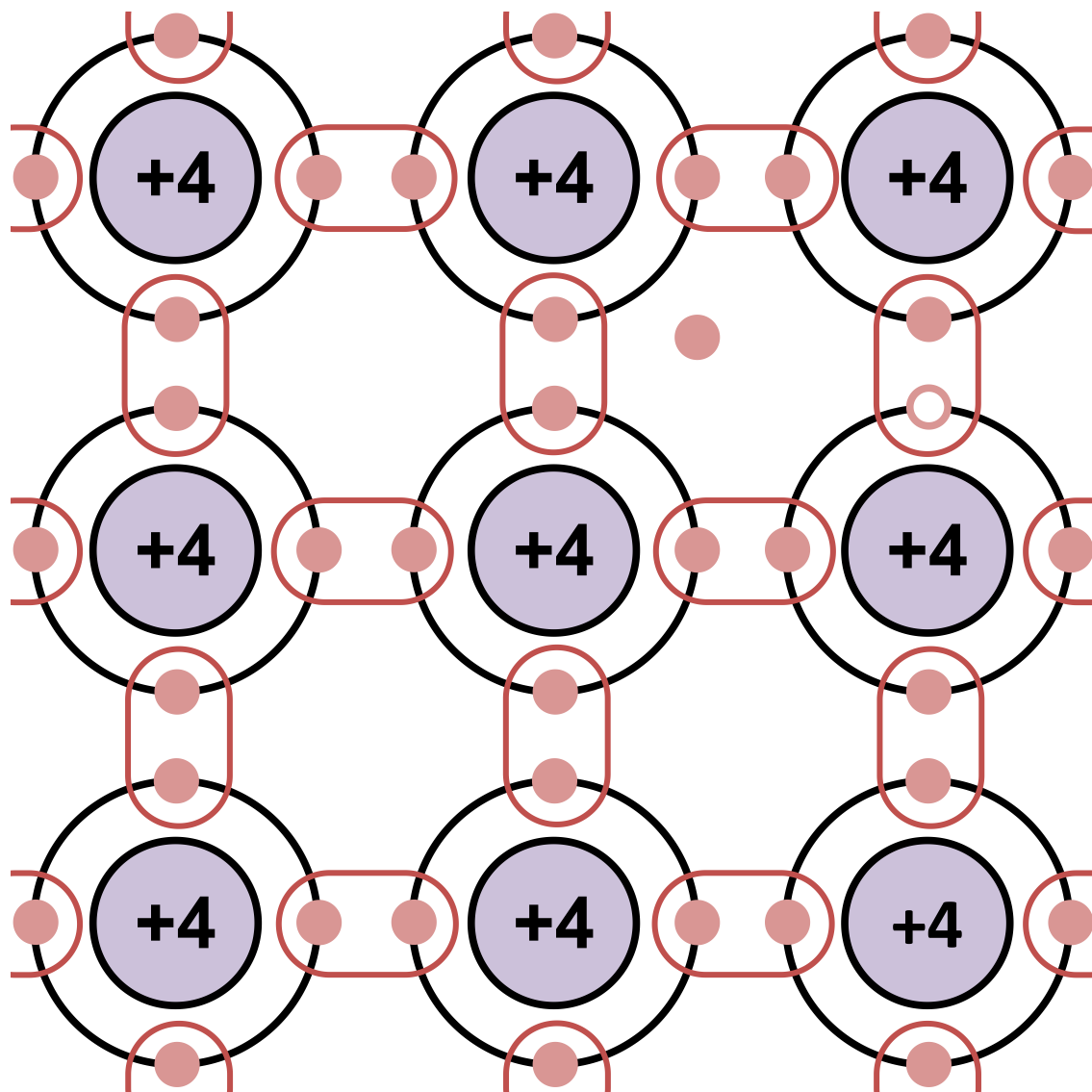


Electrons get more energy



More electron and hole pairs

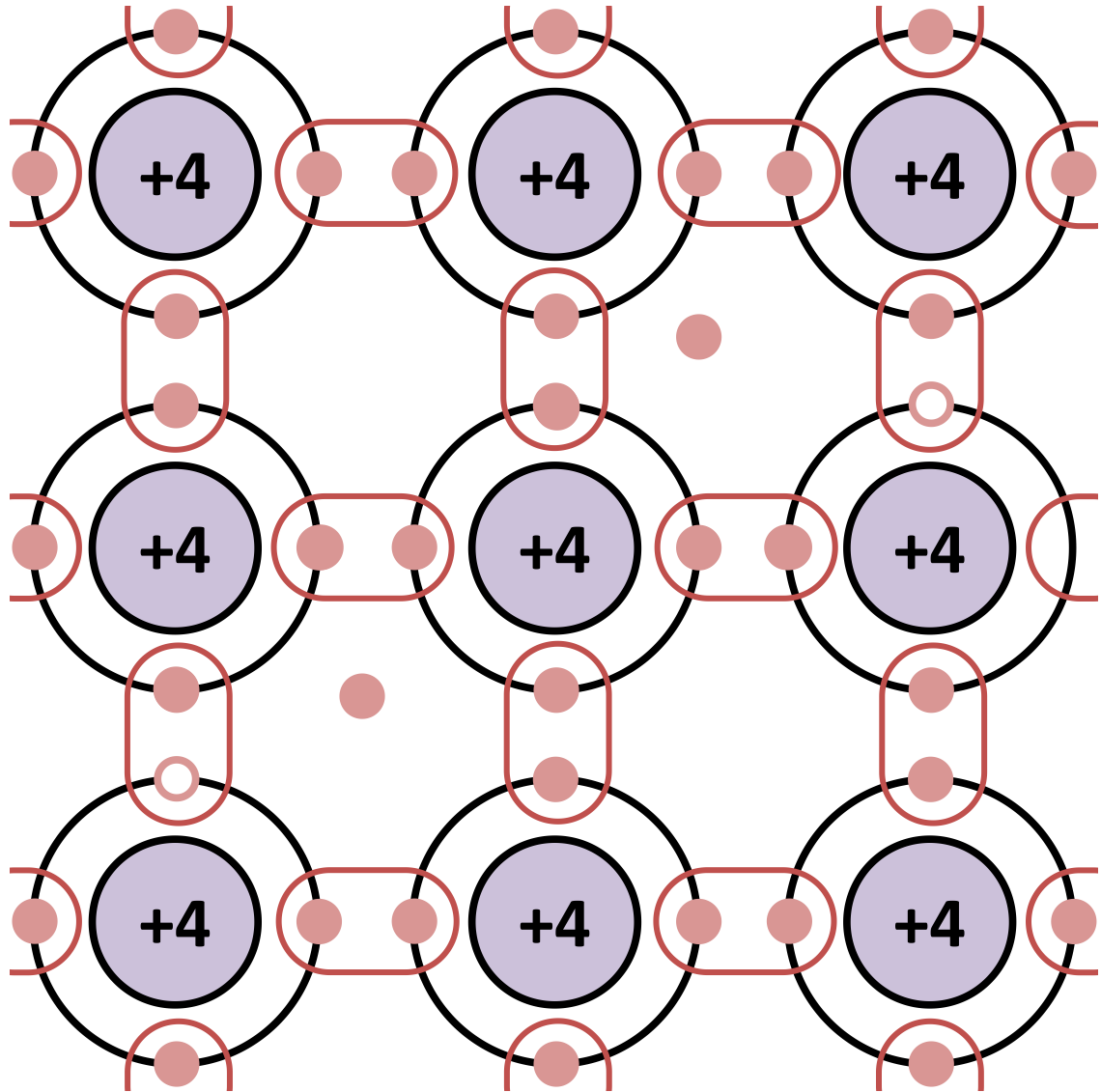
温度越高，电子空穴对数目越多



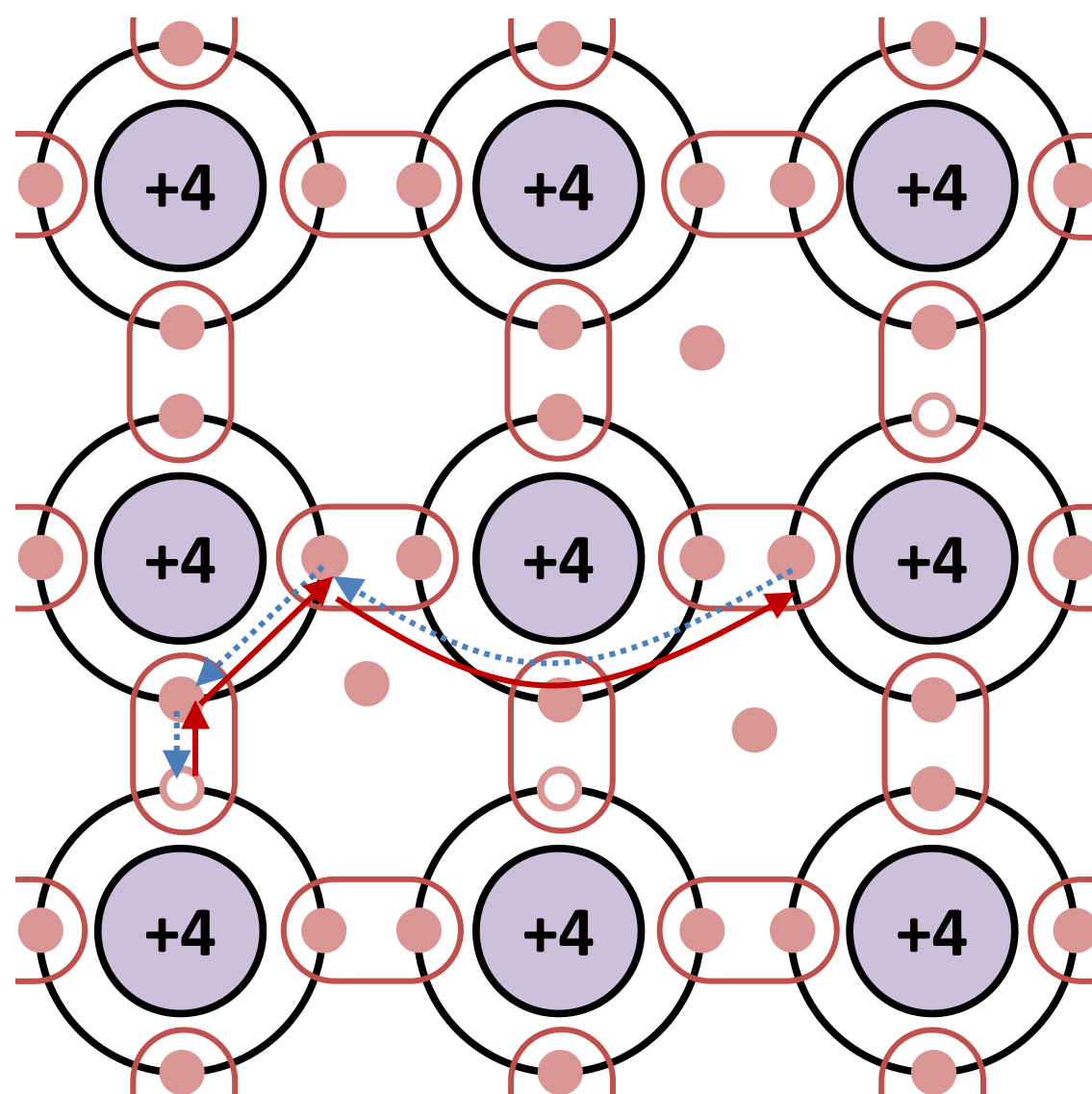
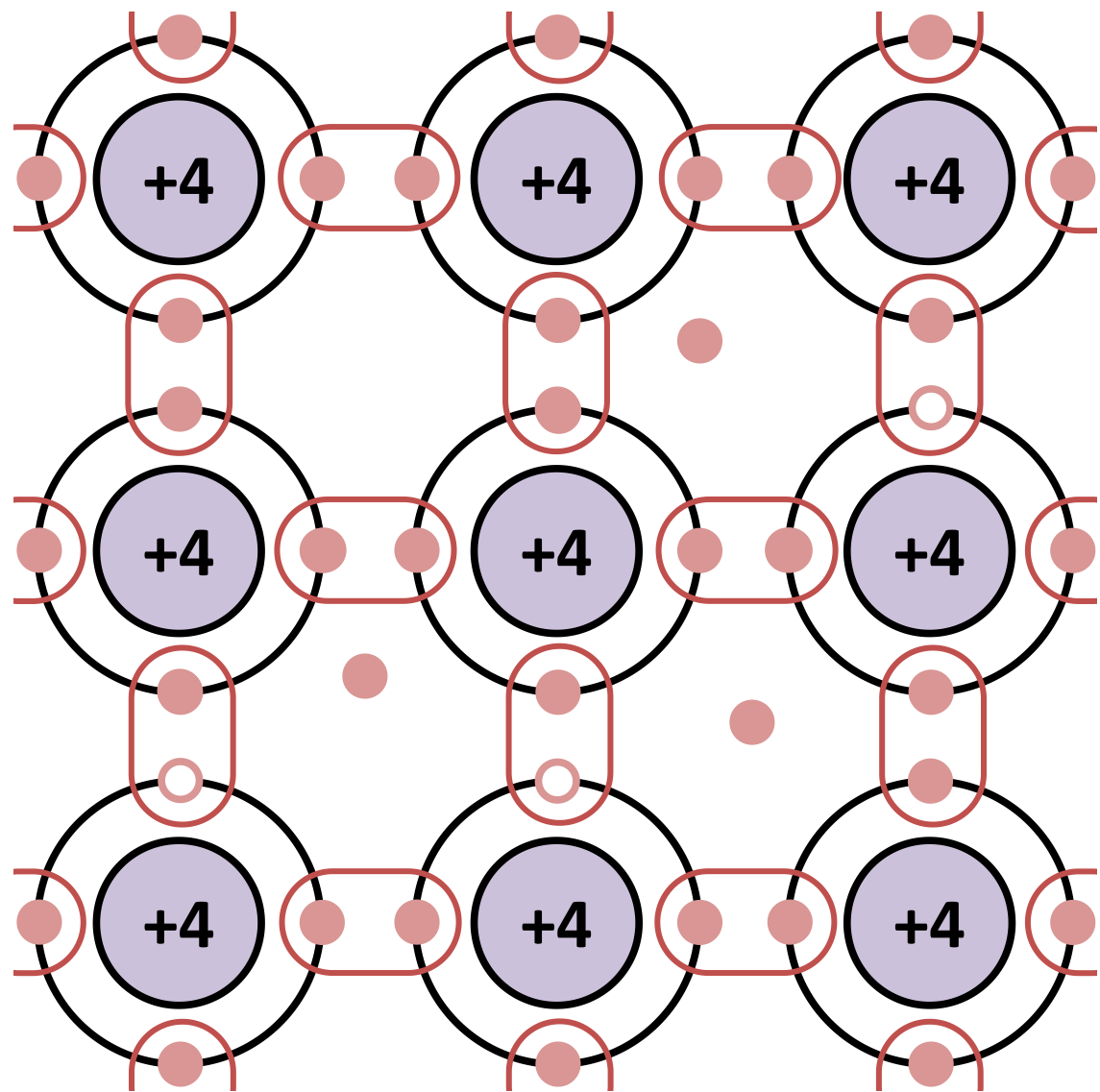
Electron and hole pair recombination 电子空穴对复合

At one time, there are both electron and hole pair excitation and recombination. They reach a dynamic equilibrium.
电子空穴对激发和复合同时存在，最终达到动态平衡。

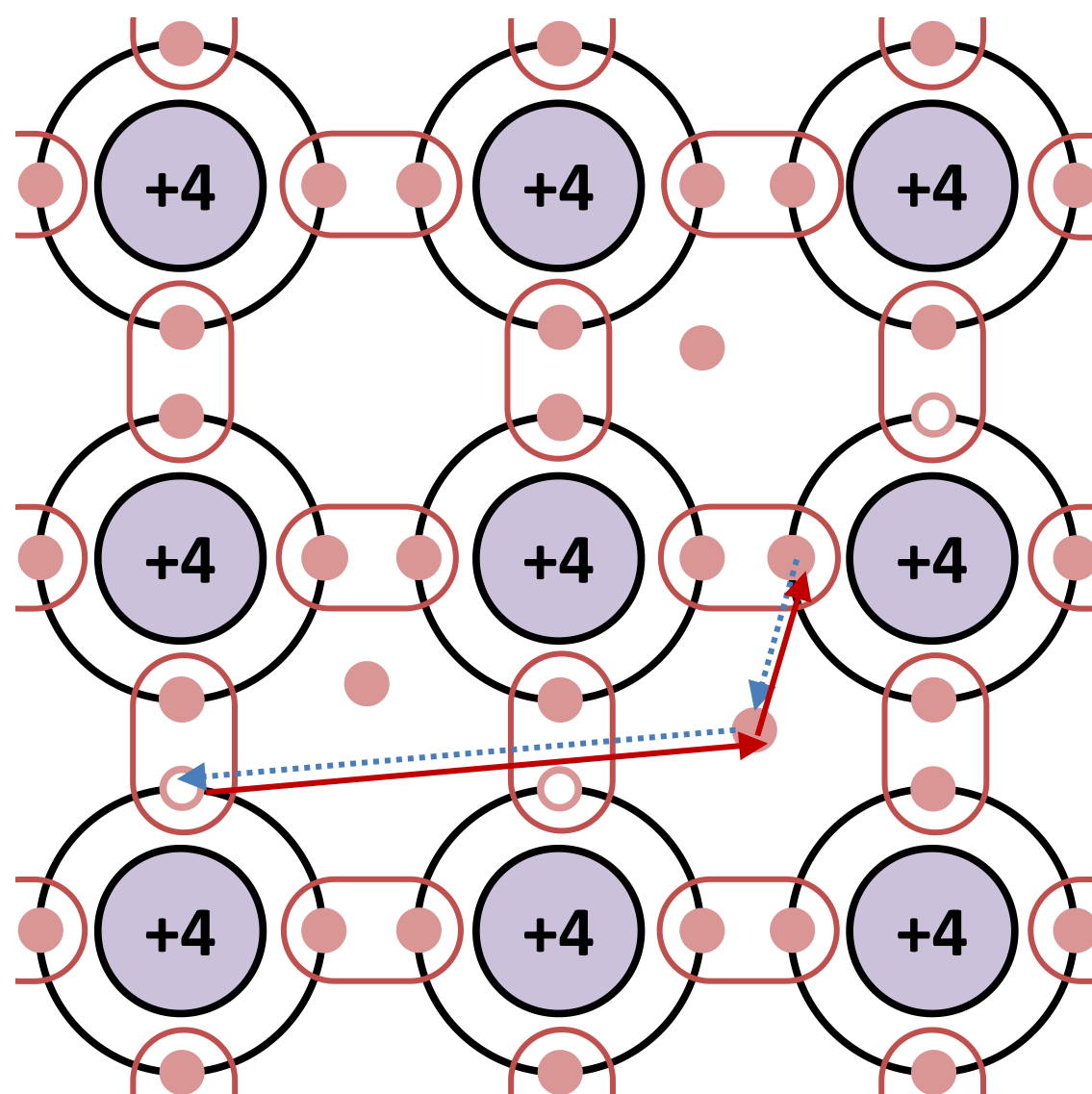
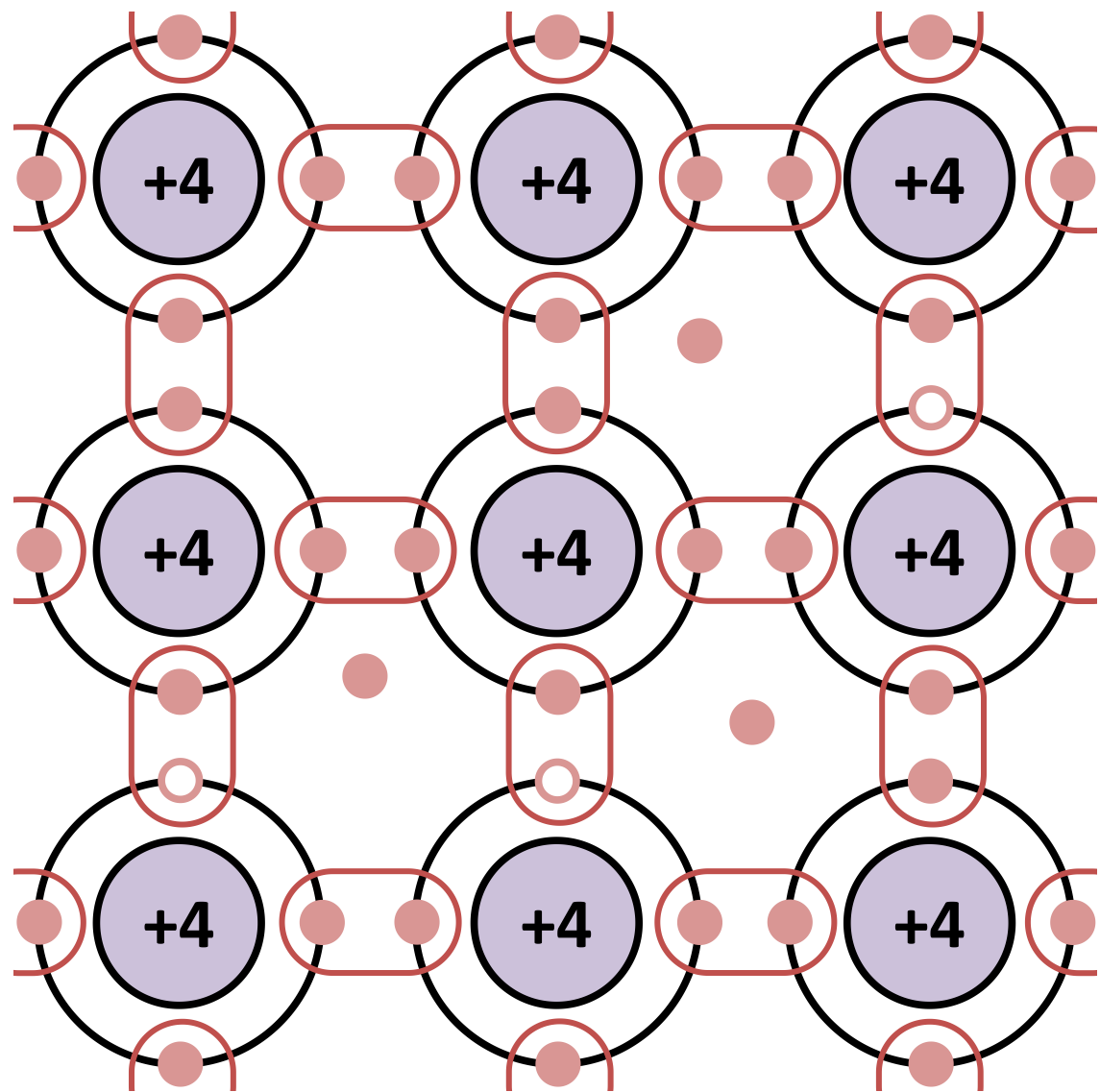
Electron concentration = Hole concentration
电子浓度=空穴浓度



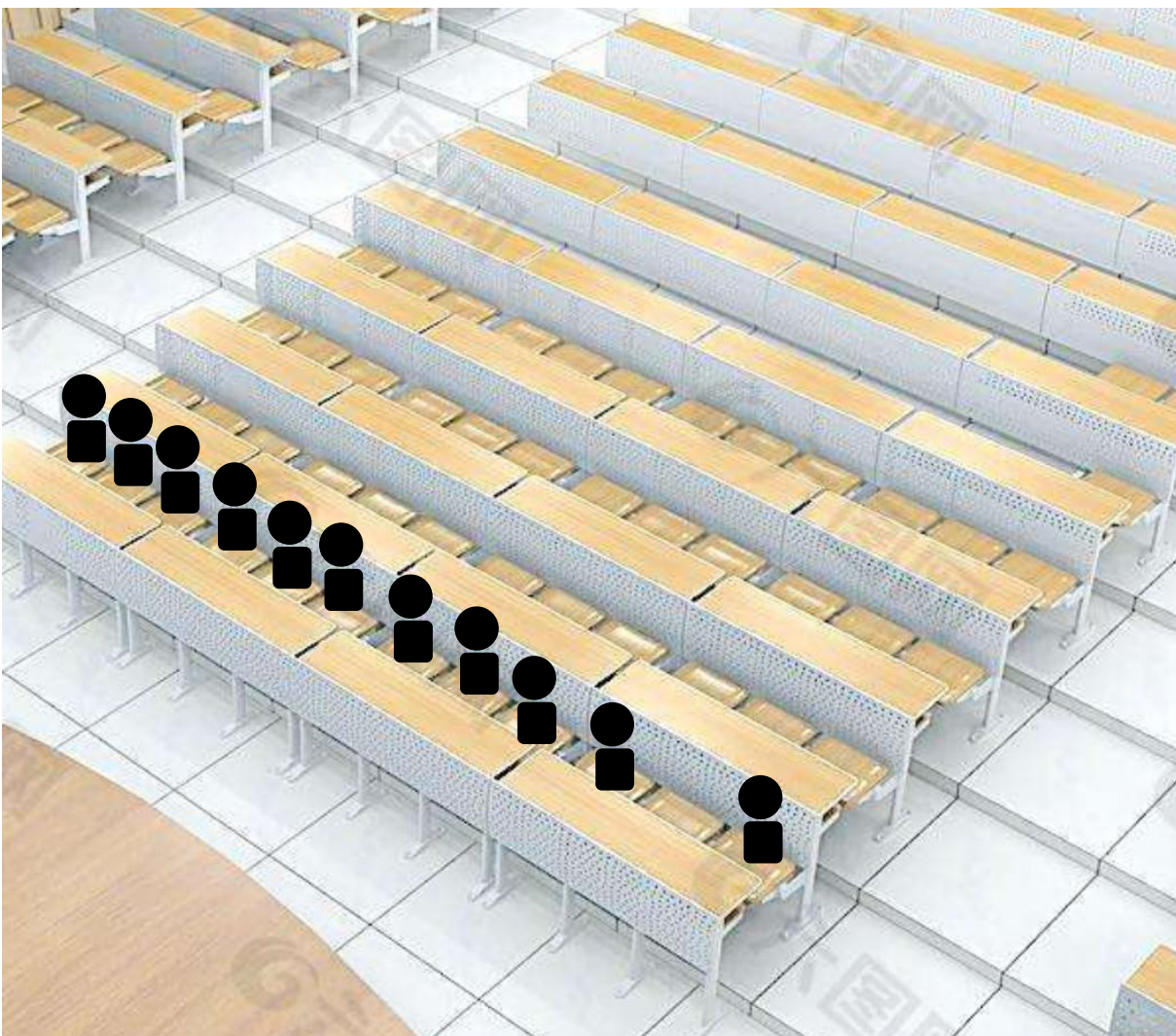
**Question: Excited electron is free.
Are holes also free?**



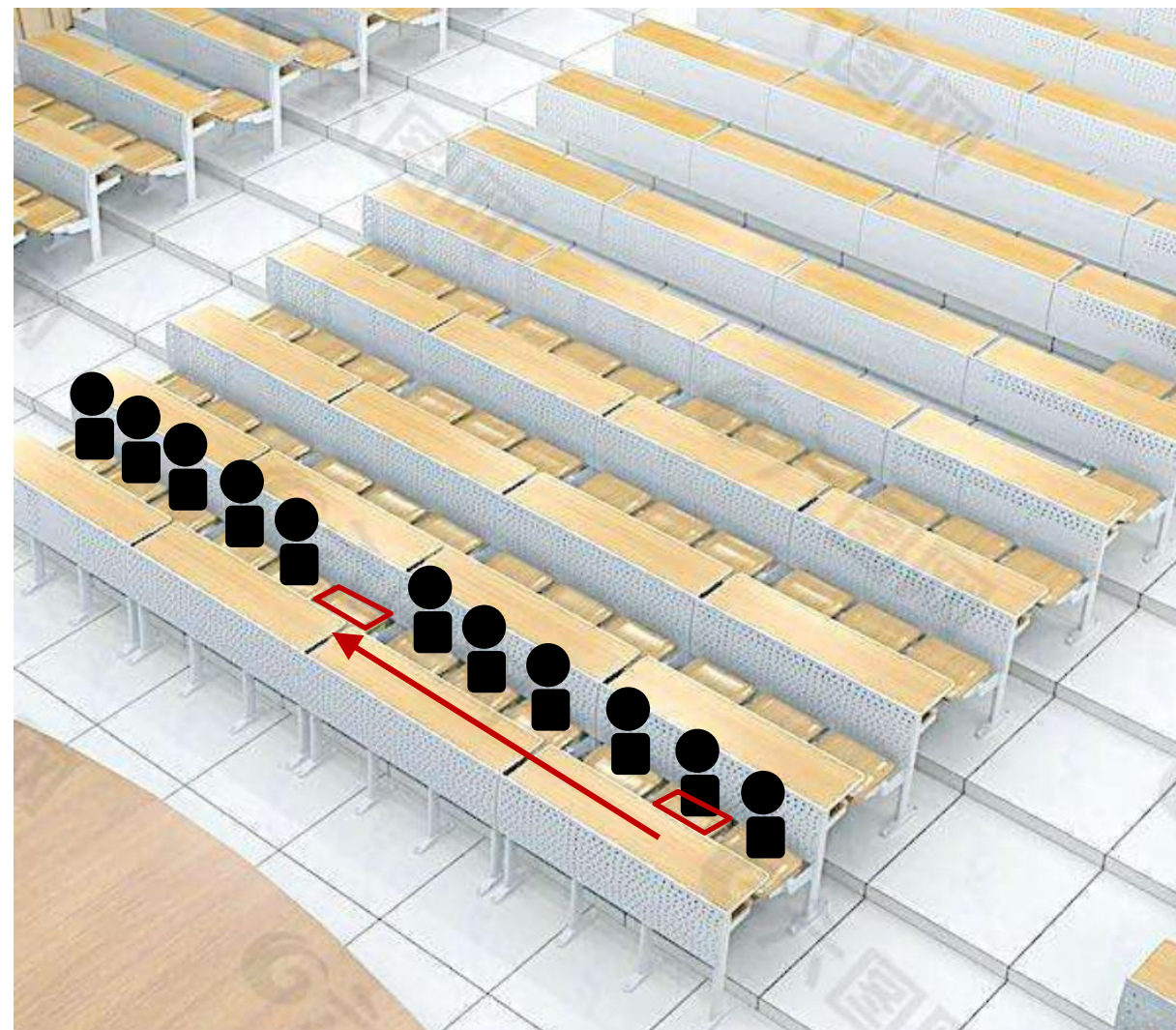
● Electron trail ○ ——— Hole trail



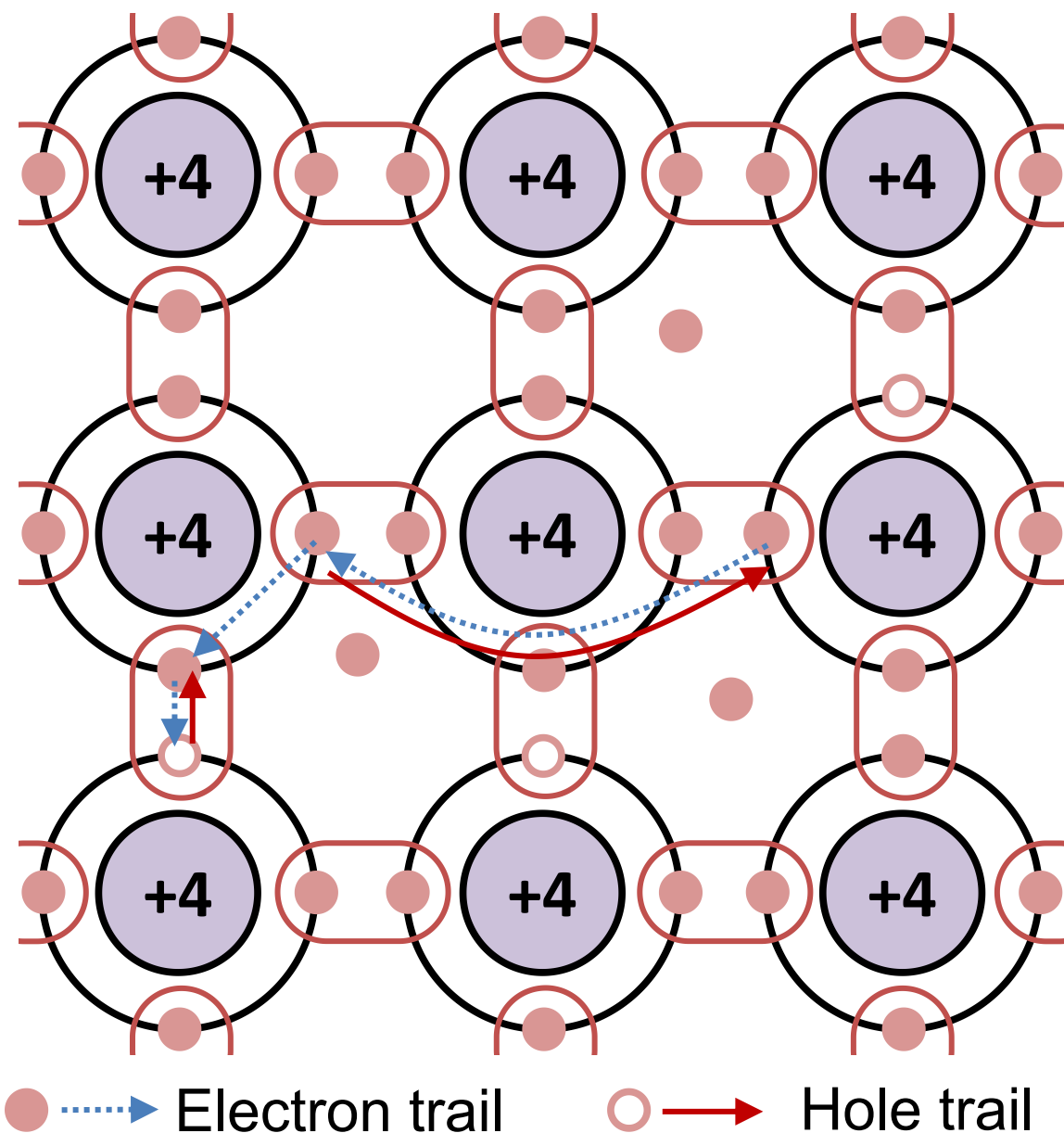
●> Electron trail ○ ———> Hole trail



 人：等效为电子
  空位：等效为空穴



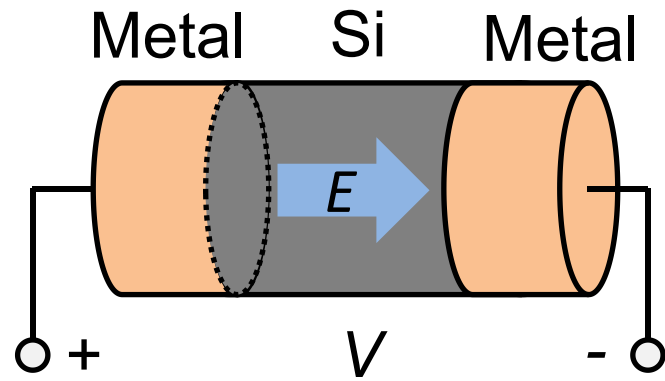
 People: electron
  Seat: hole



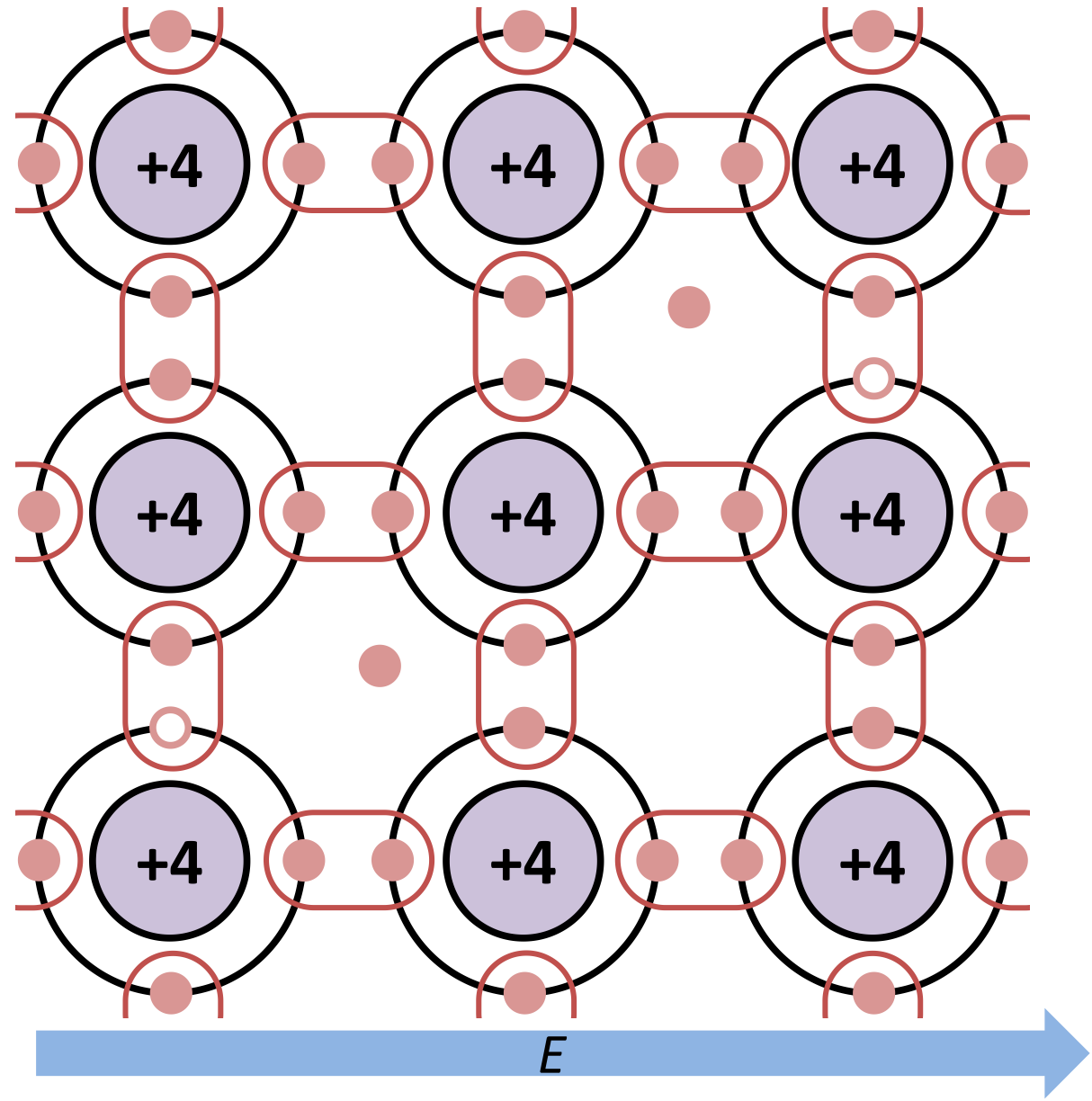
Electron excitation and recombination is equivalent to say hole is moving.
电子的激发和复合可以等效成空穴在运动。

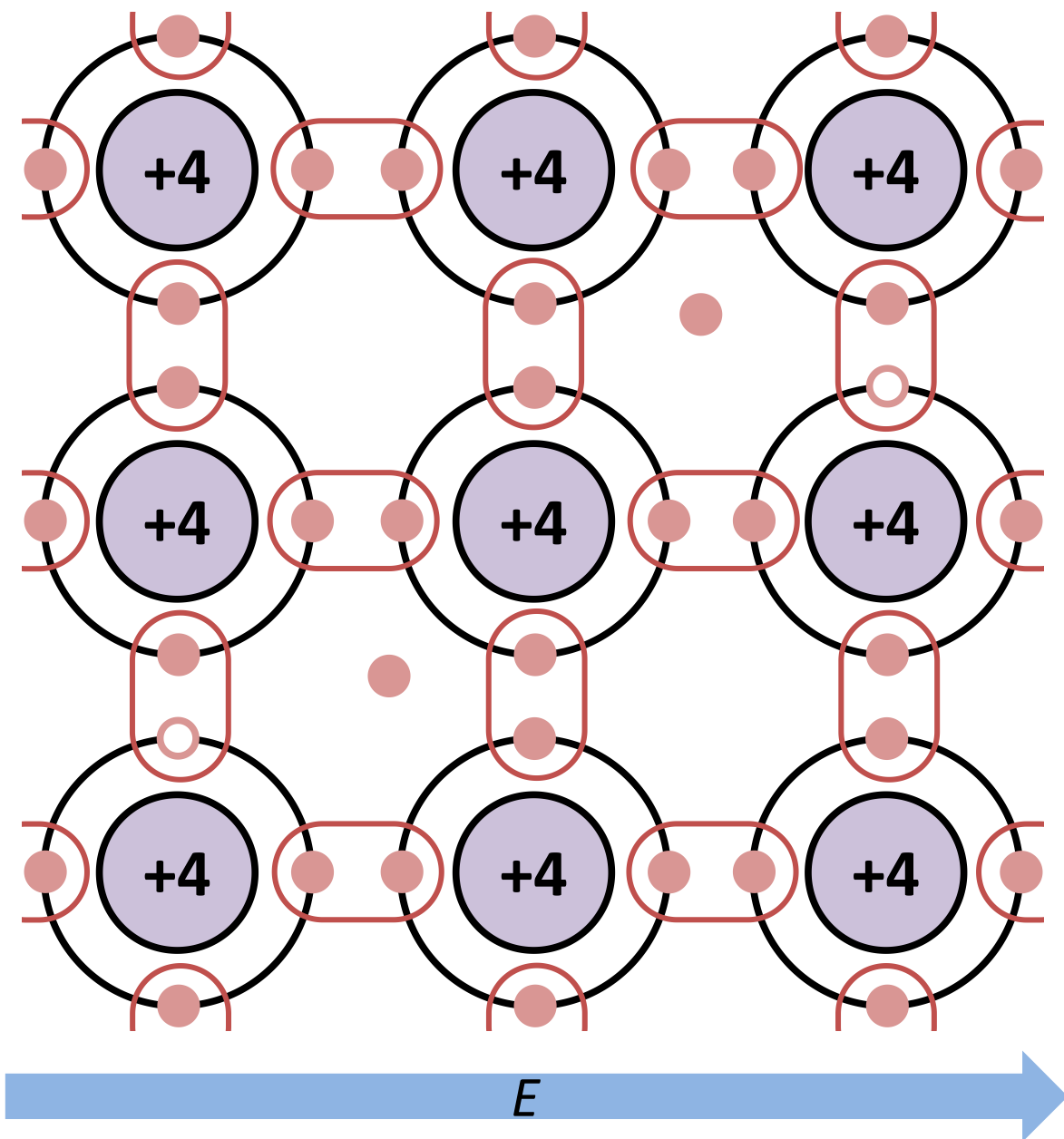
Free hole! 空穴也是自由的！

1.2 The conduction of intrinsic semiconductor 本征半导体导电

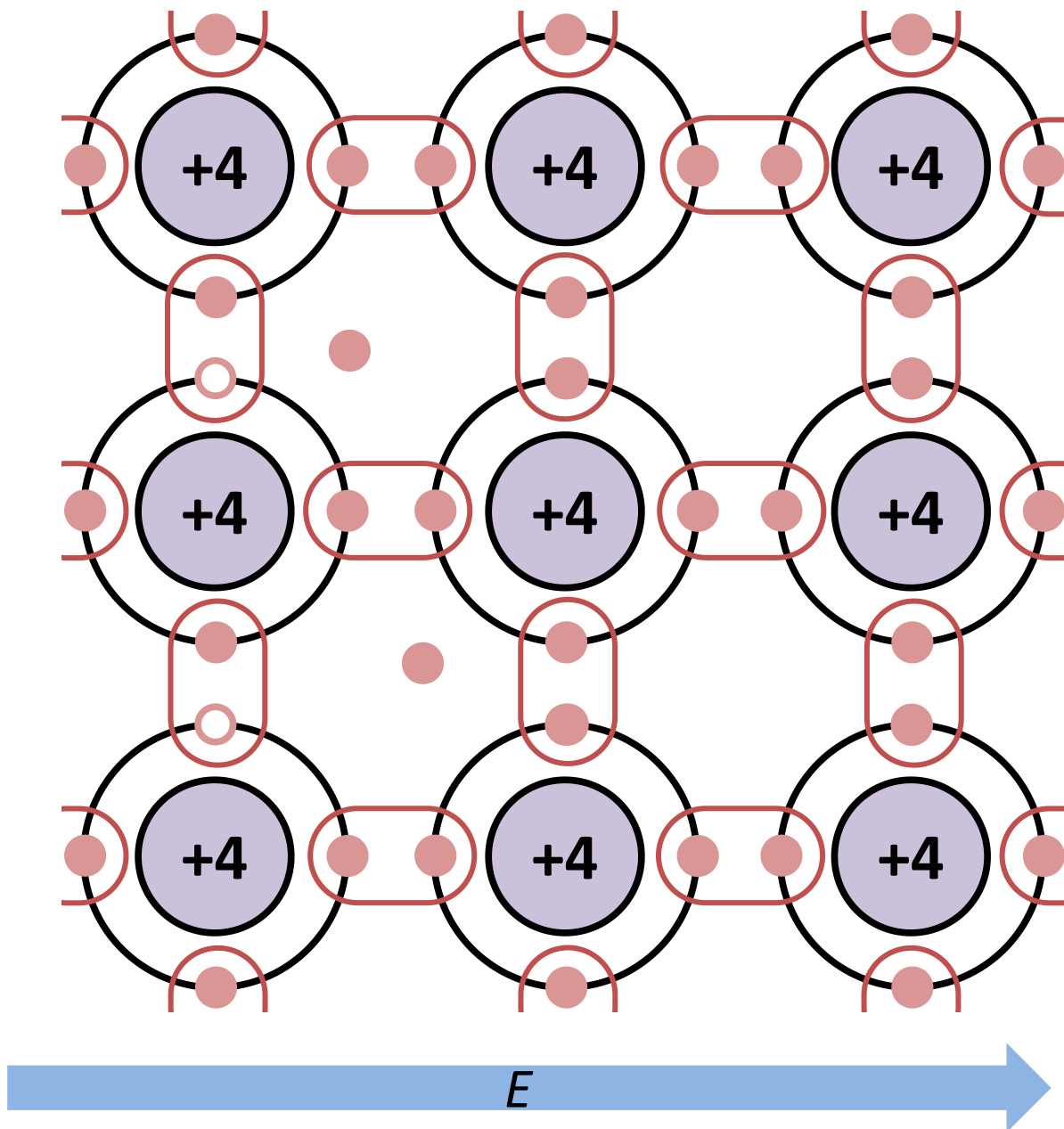


E : Electric field 电场





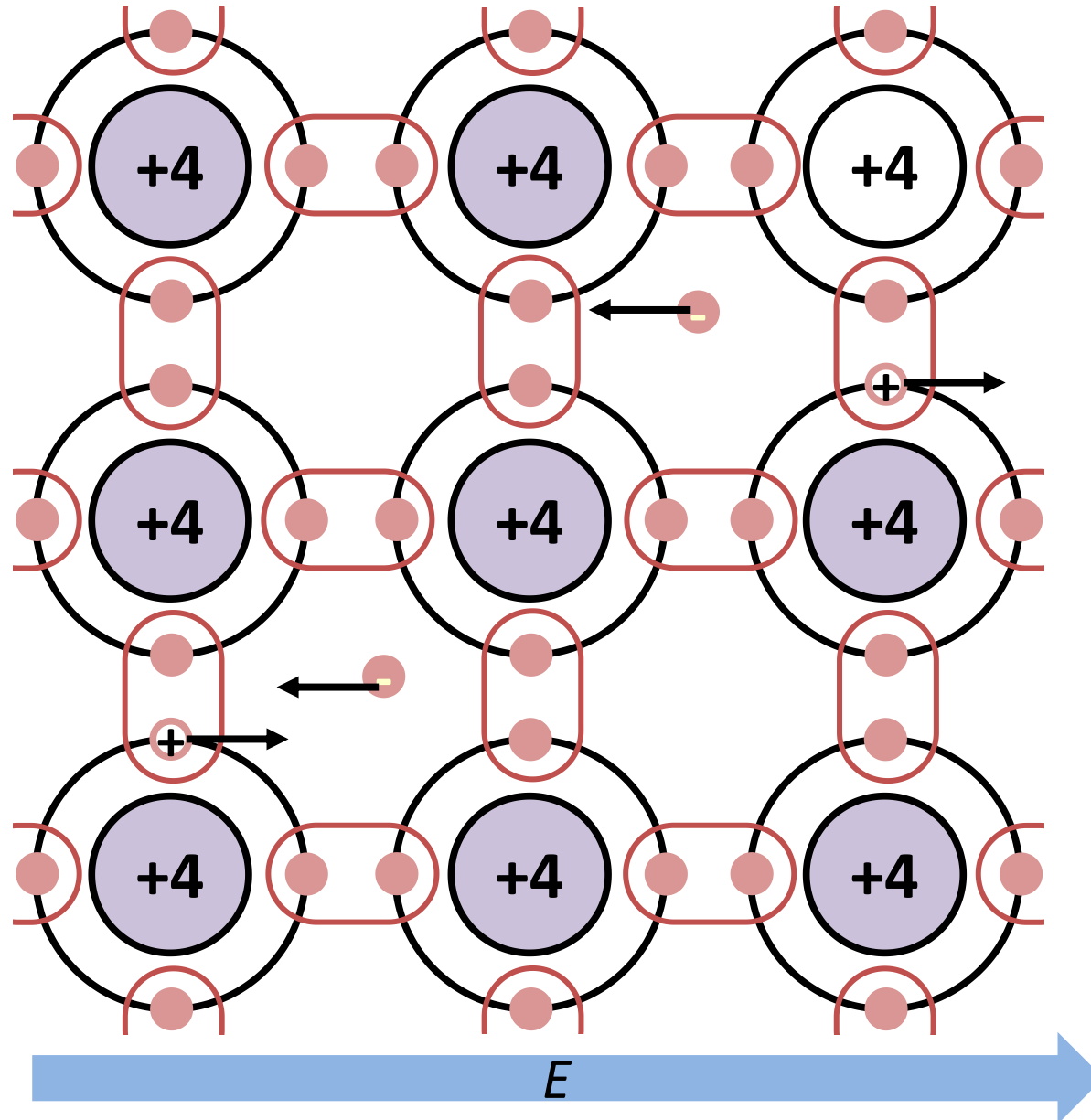
Conduction of
activated electrons
本征激发电子导电

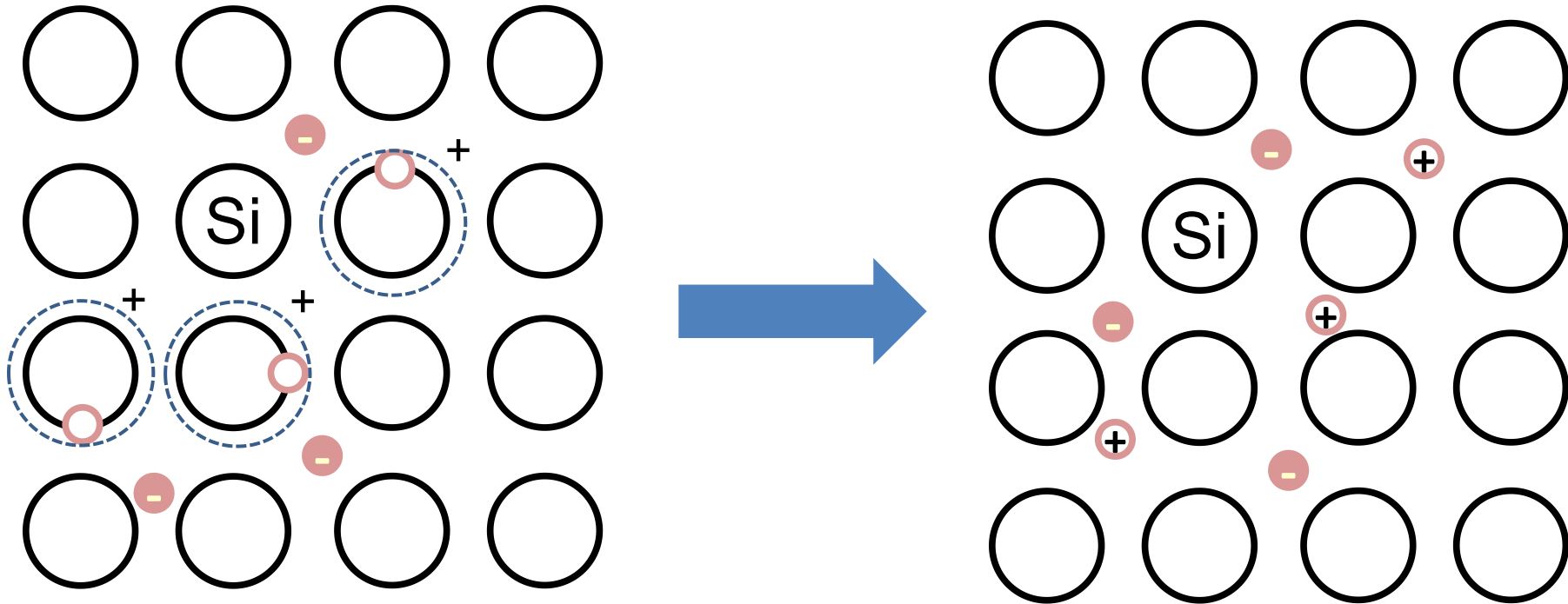


Equivalent to say holes
are moving
等效成空穴在运动

Treat holes as particles with
positive charge
可以把空穴看作一种带正电荷
的等效粒子

The conduction of semiconductor is contributed from both electrons and holes



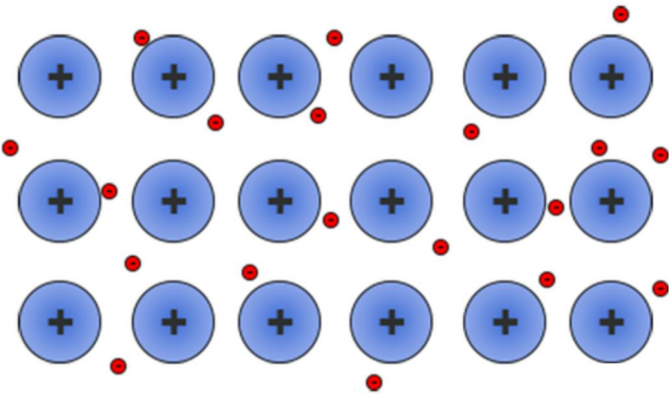


◆ **Treat holes as free particles with positive charge**
空穴可以等效成一个自由的带正电的粒子

◆ **Free electron numbers = Free hole numbers**
自由电子的个数=自由空穴的个数

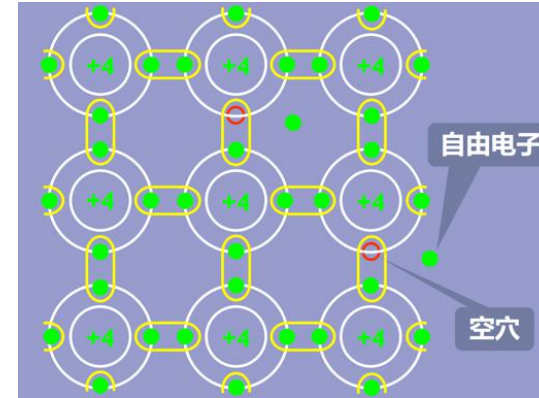
Comparison between metal and semiconductor

Metal 金属



- ◆ Only electrons, no holes
- ◆ Electrons are all free
- ◆ High conductivity

Semiconductor 半导体



- ◆ Both electrons and holes
- ◆ Activated electrons and holes are free
- ◆ Conductivity strongly depends on temperature and light excitation

Question: According to the strong temperature and light-excitation dependence of conductivity in semiconductor, please think about some applications.

Thermistor
热敏电阻/测温计

Photodetector
光电探测器



1 award point
1奖励分

1.3 Summary of intrinsic semiconductor 总结

◆ **Electrical conduction is contributed from both:**

- Electrons with negative charges 
- Holes with positive charges 

◆ **Electrical conduction of intrinsic semiconductor.**

Low conductivity 电导率较低.

Strong temperature-dependence 温度效应明显.

Higher temperature, more electron-hole pairs, and higher conductivity.

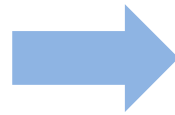
Suitable for thermistor and photoresistor.

Conductivity of intrinsic semiconductor is too low for diode and transistor applications.

本征半导体电导率较低，不适合用在二极管和晶体管中。

Q: How to improve the conductivity of semiconductors?

Intrinsic semiconductor



Doped semiconductor

引用掺杂半导体

1.1.2 Doped/Extrinsic semiconductor 掺杂半导体

Substitute intrinsic Si/Ge atoms with foreign atoms.

在本征半导体硅或锗中掺入微量的其它适当元素后所形成的半导体

Depends on dopant (掺杂物), doped semiconductors can be categorized as

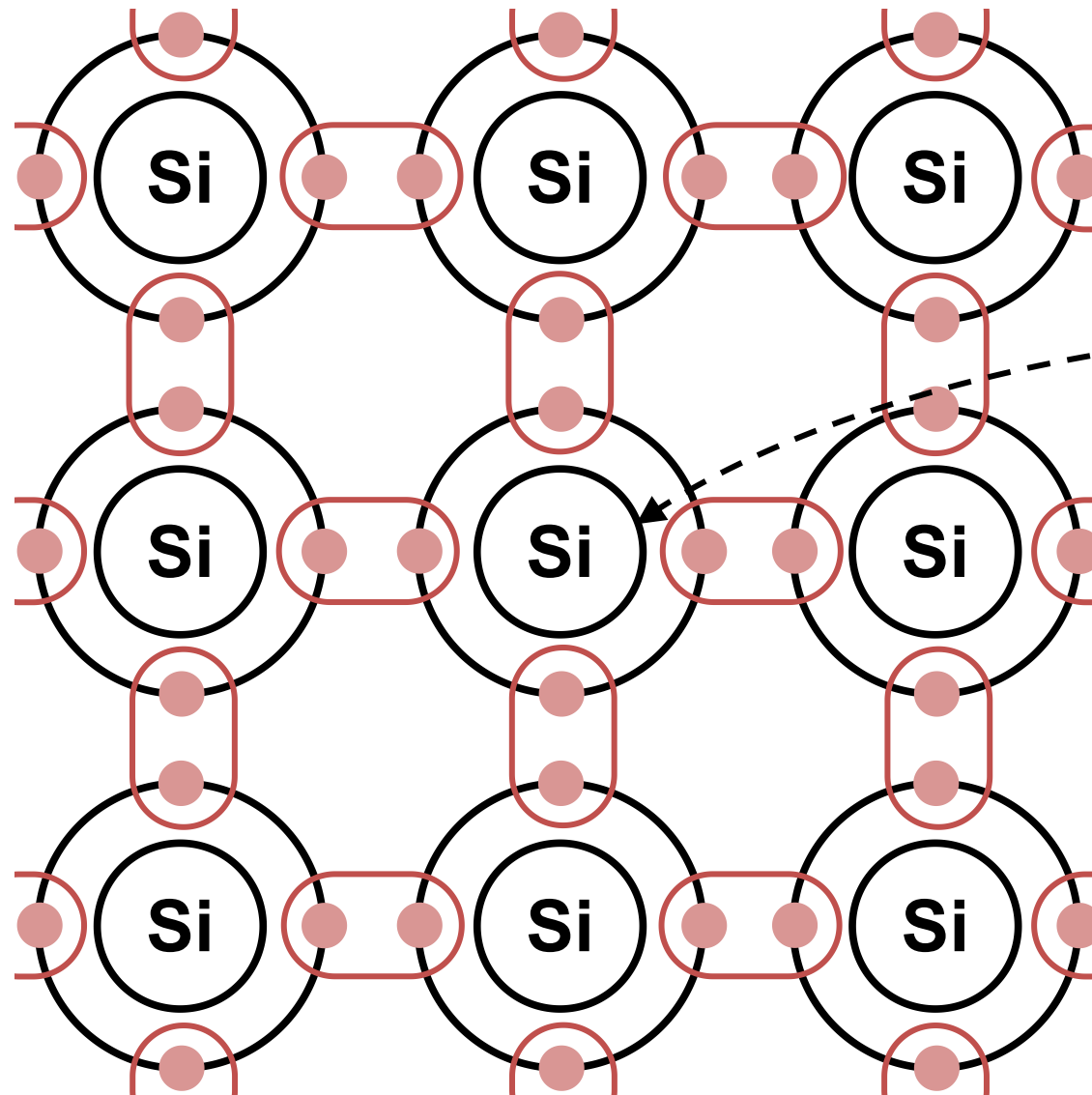
N-type semiconductor N型半导体

N represents “Negative”. Electrons with negative charge are major conducting carriers.

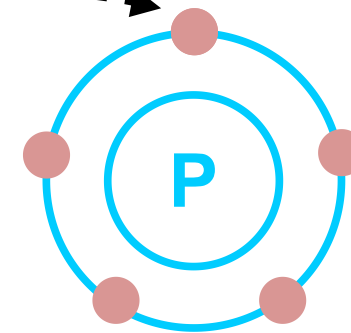
P-type semiconductor P型半导体

P represents “Positive”. Holes with positive charge are major conducting carriers.

N-type semiconductor

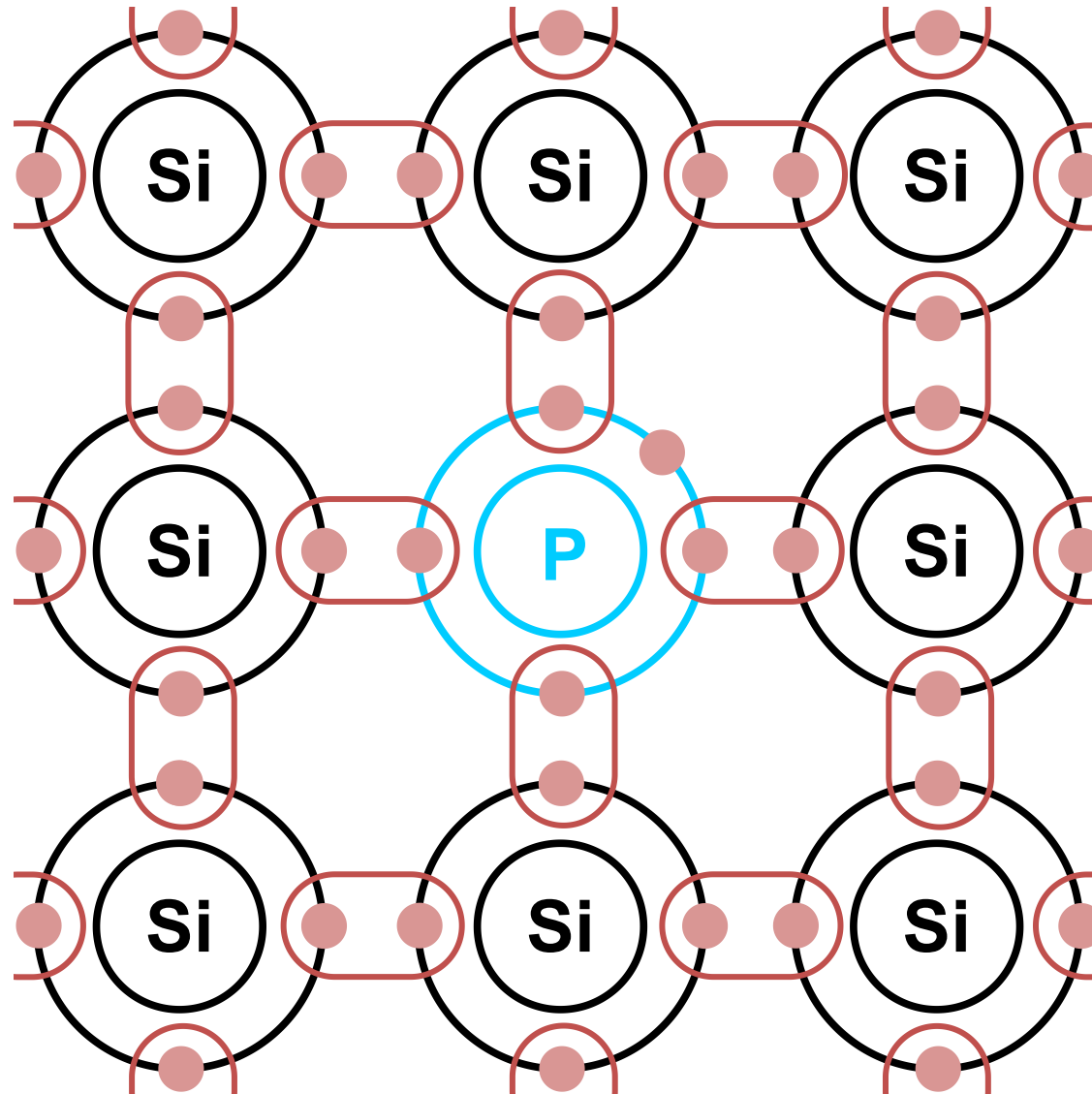


Dopant:
Phosphorus(磷)
and Arsenide(砷)

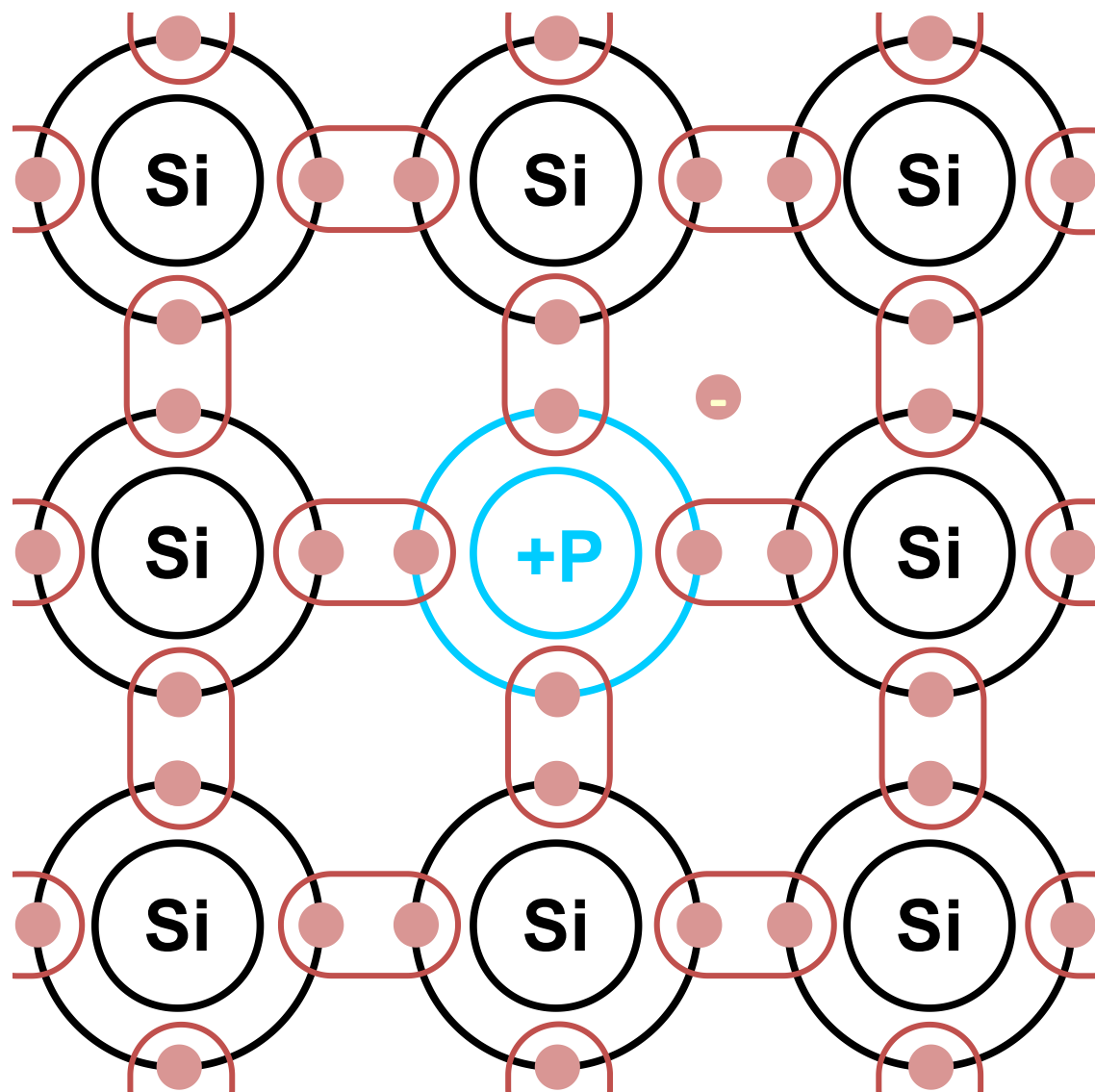


5 electrons in
outmost shell

Dopant: Phosphorus and Arsenide



Dopant: Phosphorus and Arsenide



One more free electron and one positive ion

Electron number > hole number

Electron: major carriers

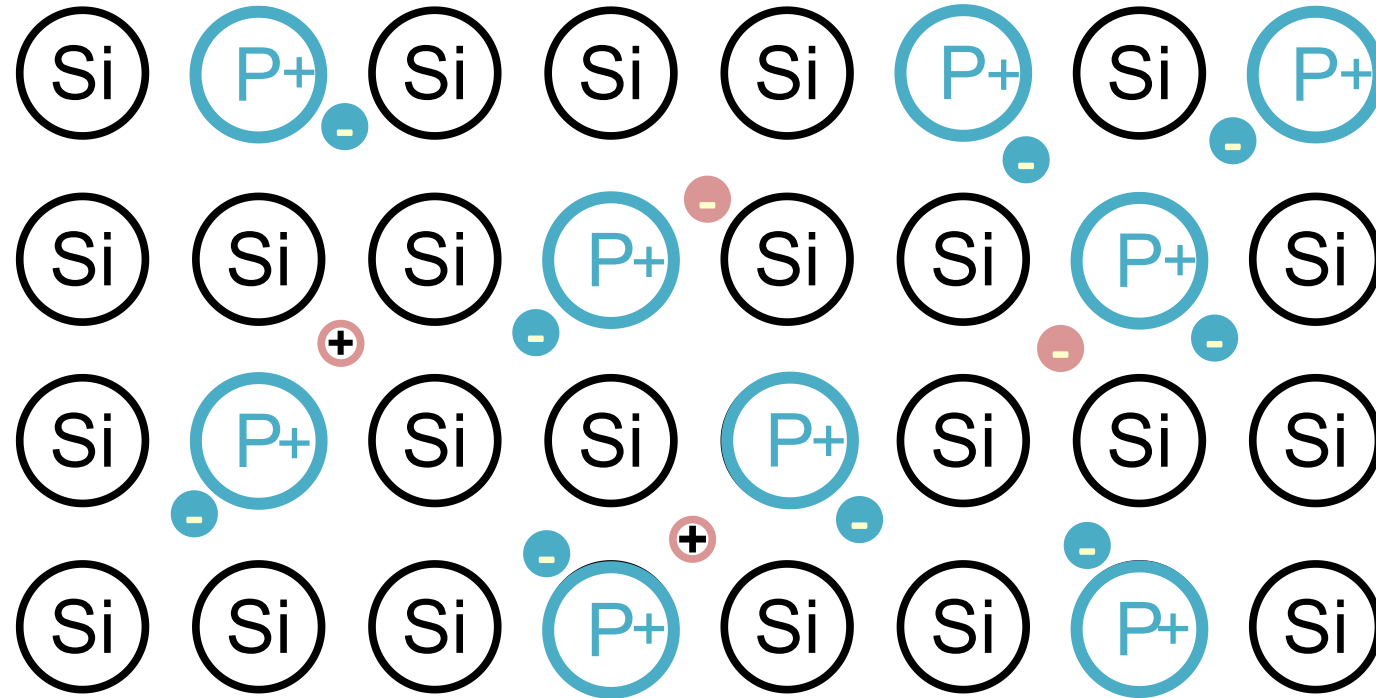
电子:多数载流子

Hole: minor carriers

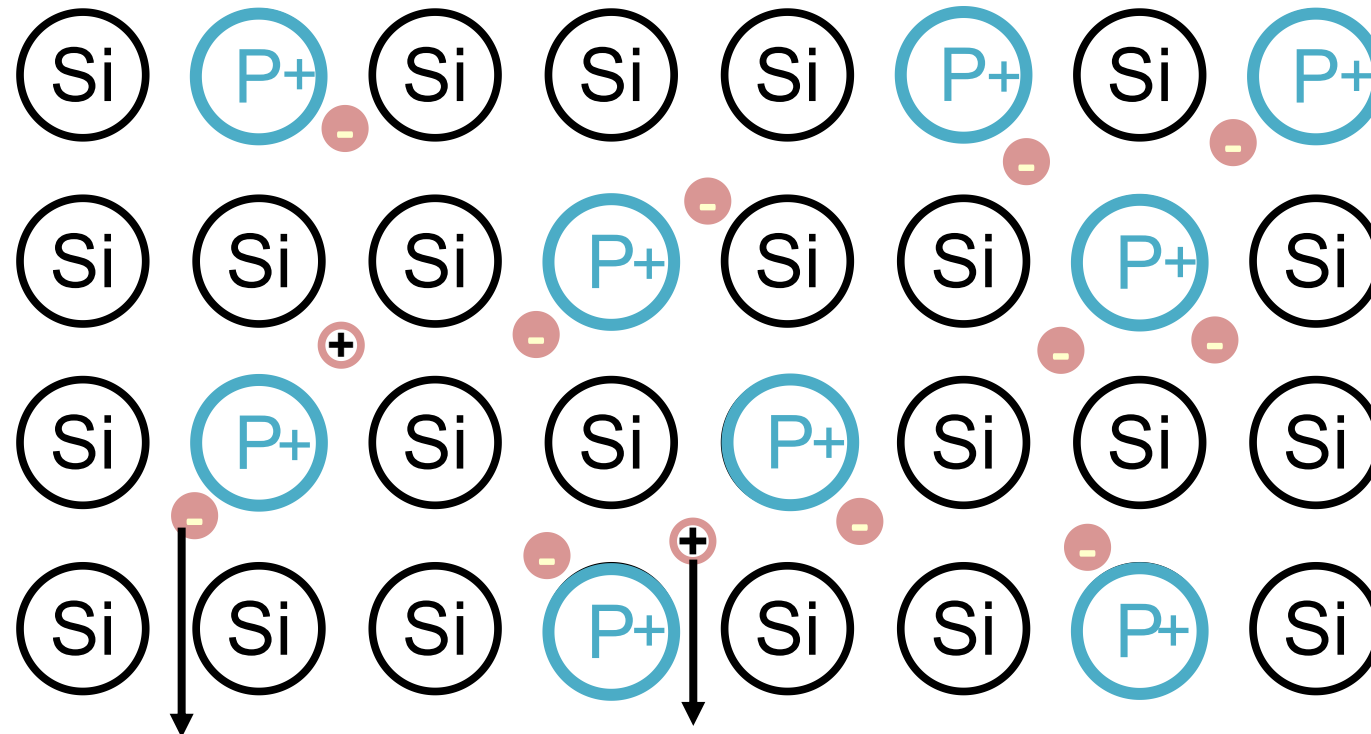
空穴:少数载流子

P and As: **donor** 施主杂质

N-type semiconductor



N-type semiconductor



Major carriers: electrons

Minor carriers: hole

Q: What's the relation between N_{e-} , N_{h+} and N_P ?

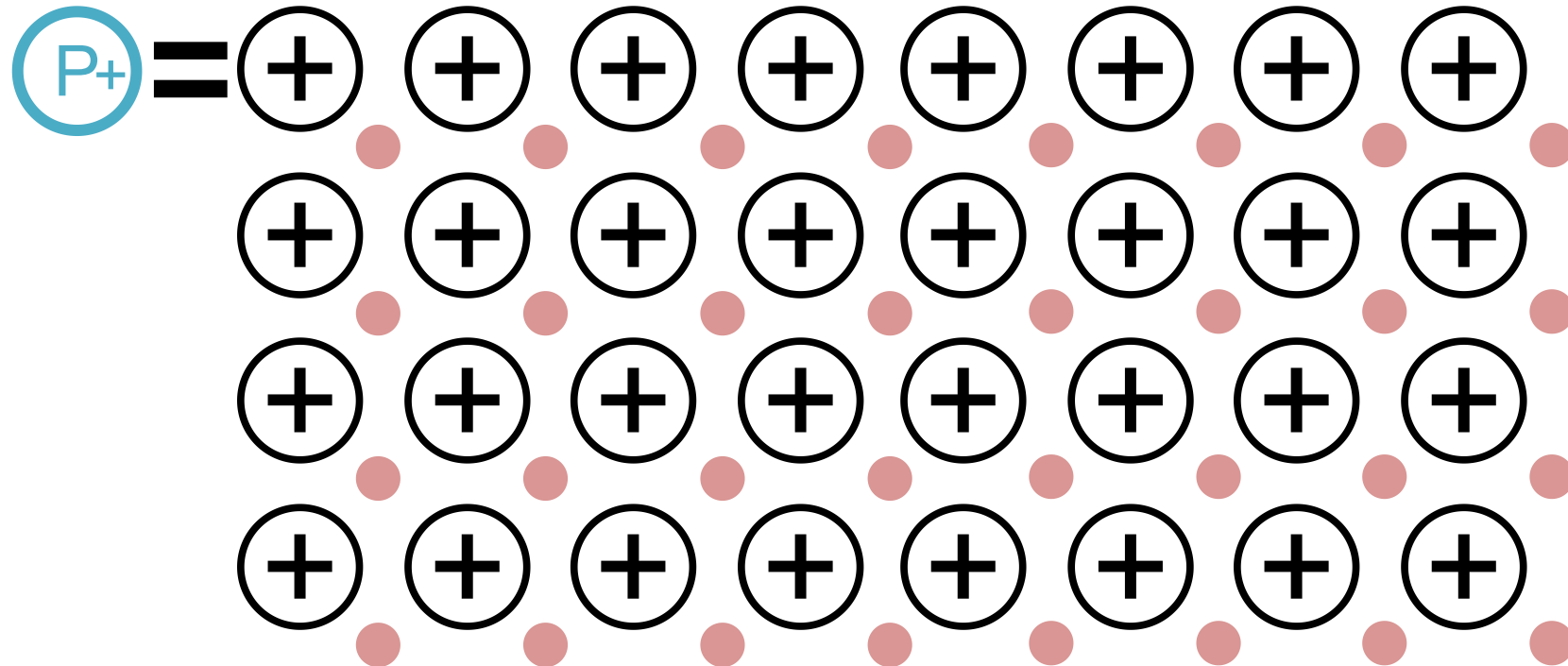
In heavily doped N-type semiconductor:

高浓度掺杂

$$\left\{ \begin{array}{l} N_{e-} \gg N_{h+} \\ N_{e-} \approx N_P \end{array} \right.$$

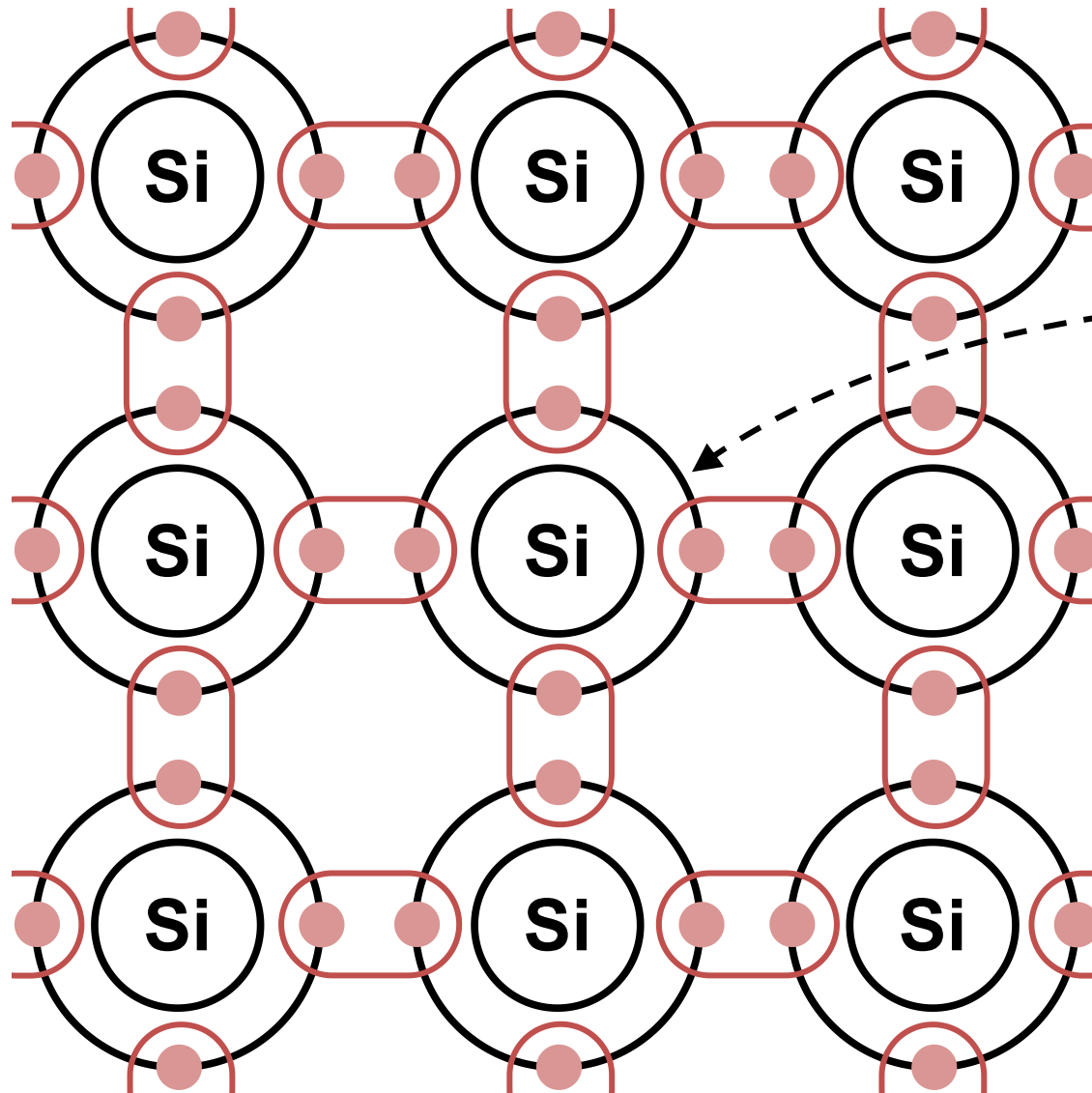
Schematic for N-type semiconductor

Donor ions 施主杂质正离子

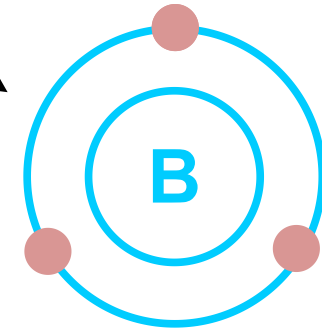


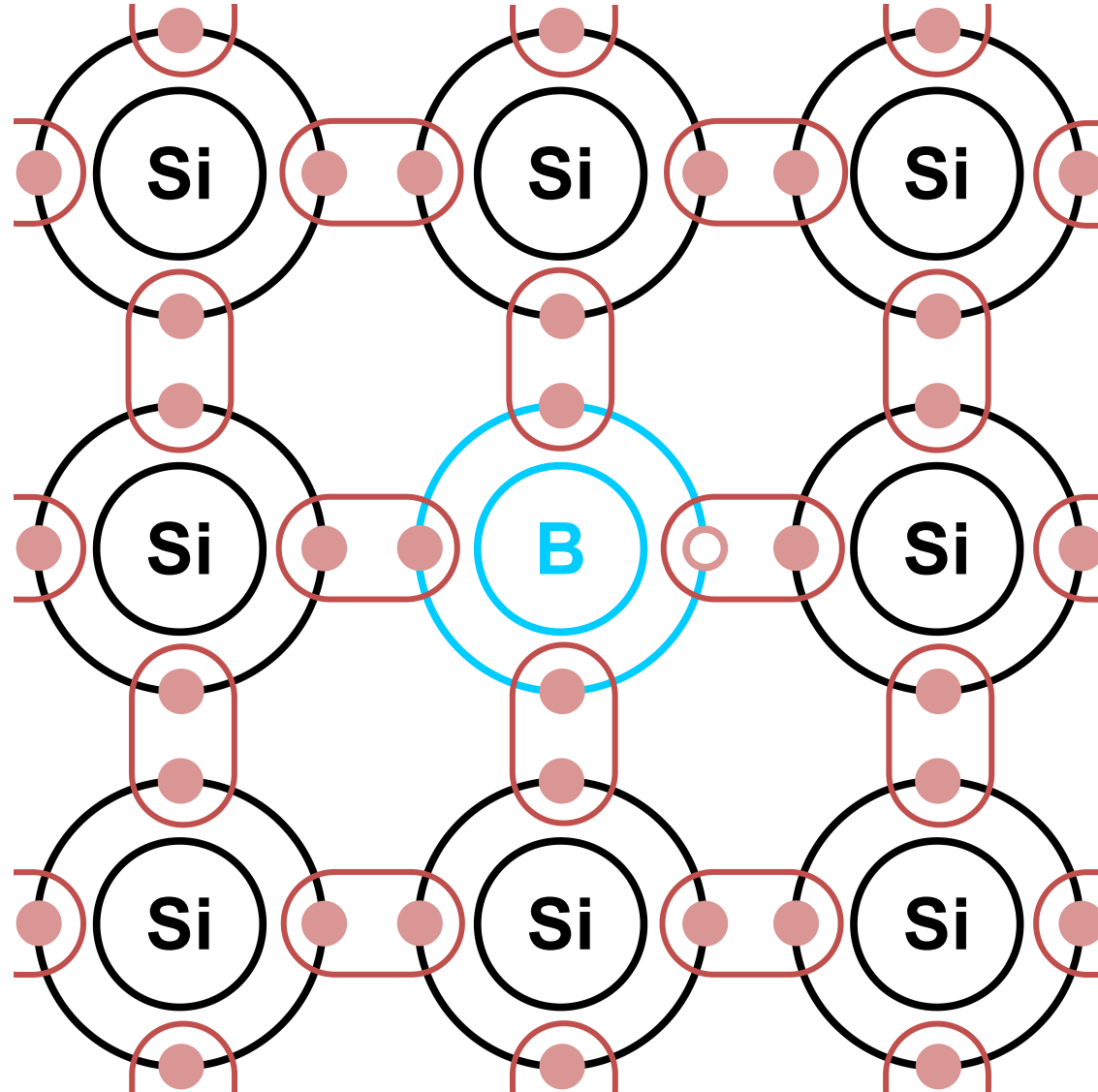
The schematic neglects Si atoms and holes
简图中硅原子和少量空穴没有标识出来

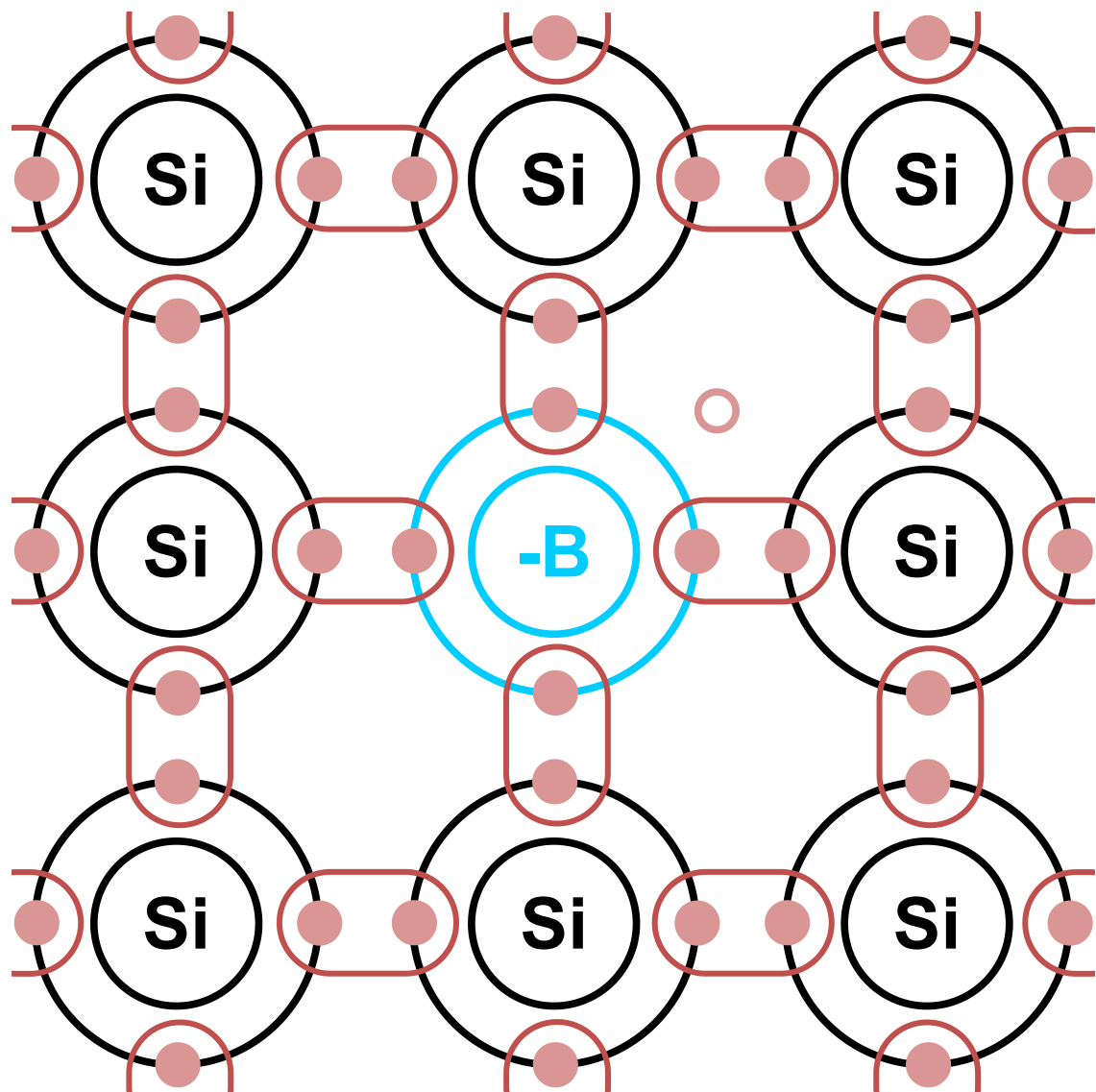
P-type semiconductor



Dopant: Boron(硼)







One more free hole and one negative ion

Hole number > electron number

Hole: major carriers

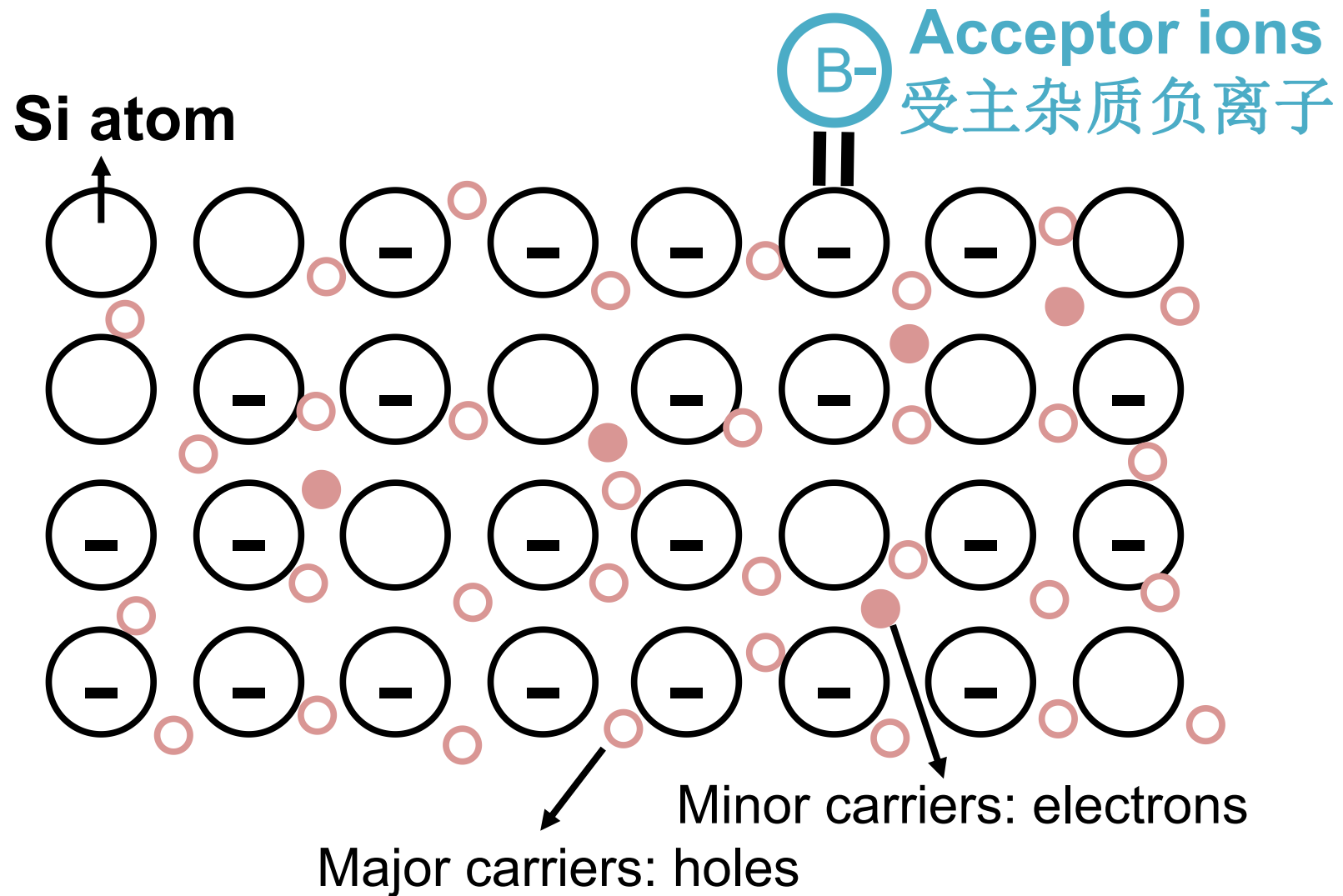
空穴:多数载流子

Electron: minor carriers

电子:少数载流子

B: Acceptor 受主杂质

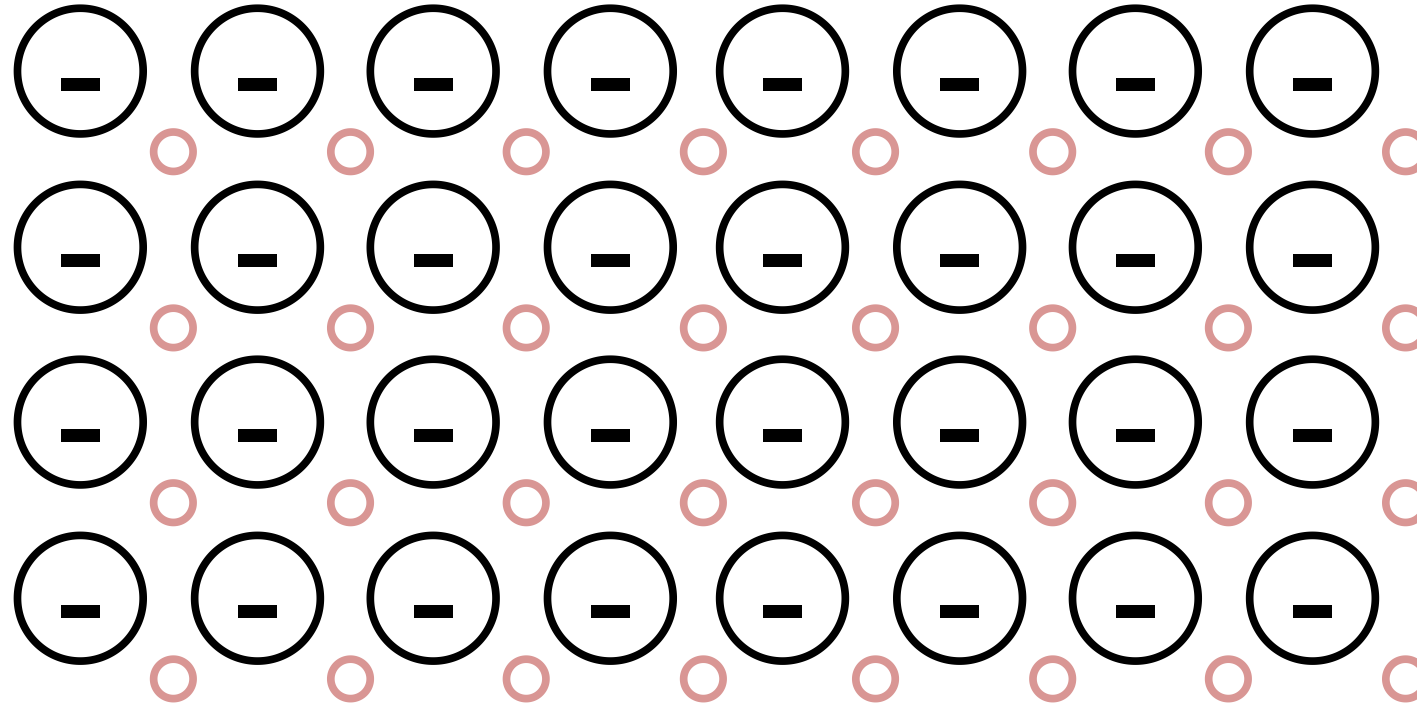
P-type semiconductor



In heavily doped P-type semiconductor:

$$\left\{ \begin{array}{l} N_{h+} \gg N_{e-} \\ N_{h+} \approx N_B \end{array} \right.$$

Schematic 示意简图 for P-type semiconductor



The schematic neglects Si atoms and electrons
简图中硅原子和少量电子没有标识出来

Summary

□ N-type semiconductor N型半导体

Electron donor: 磷(P), 砷(As): 5 electrons in outmost shell

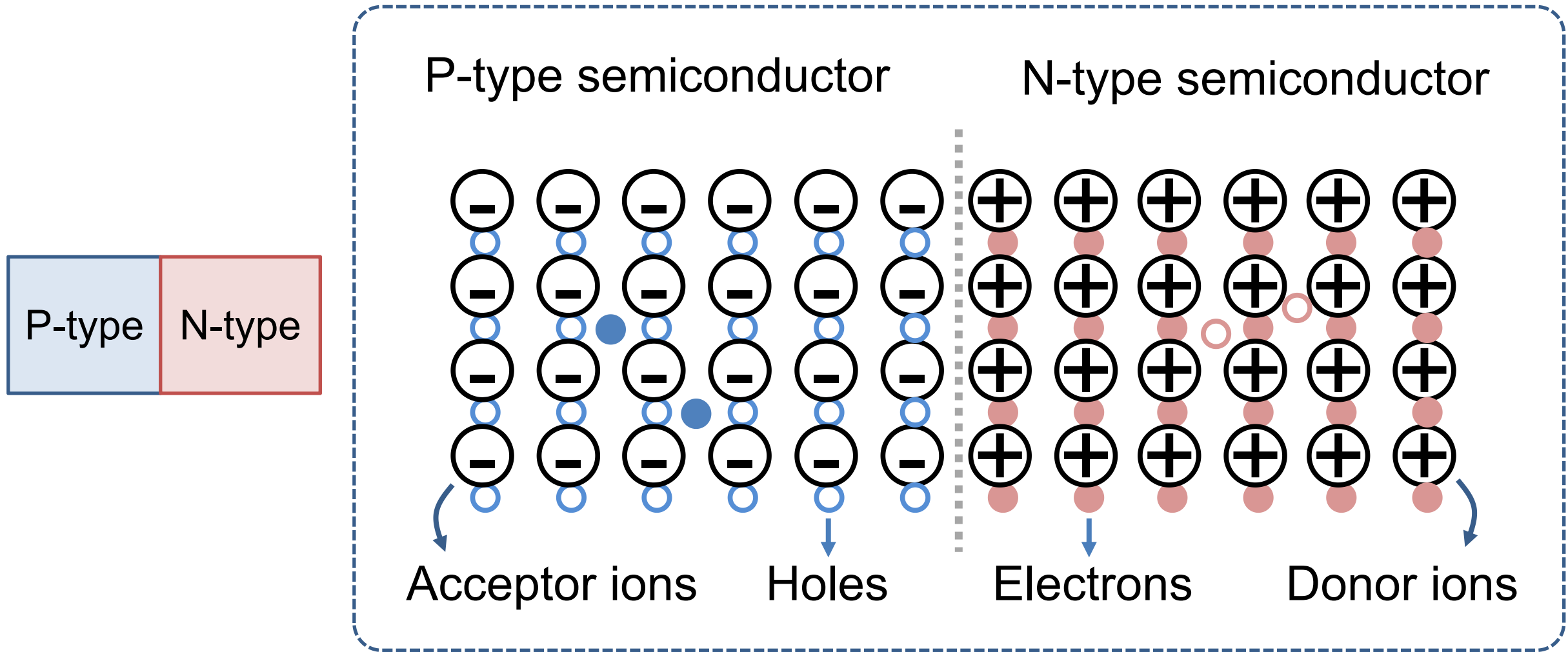
Electrons are majority carriers, holes are minority carriers
电子是多数载流子, 空穴是少数载流子

□ P-type semiconductor P型半导体

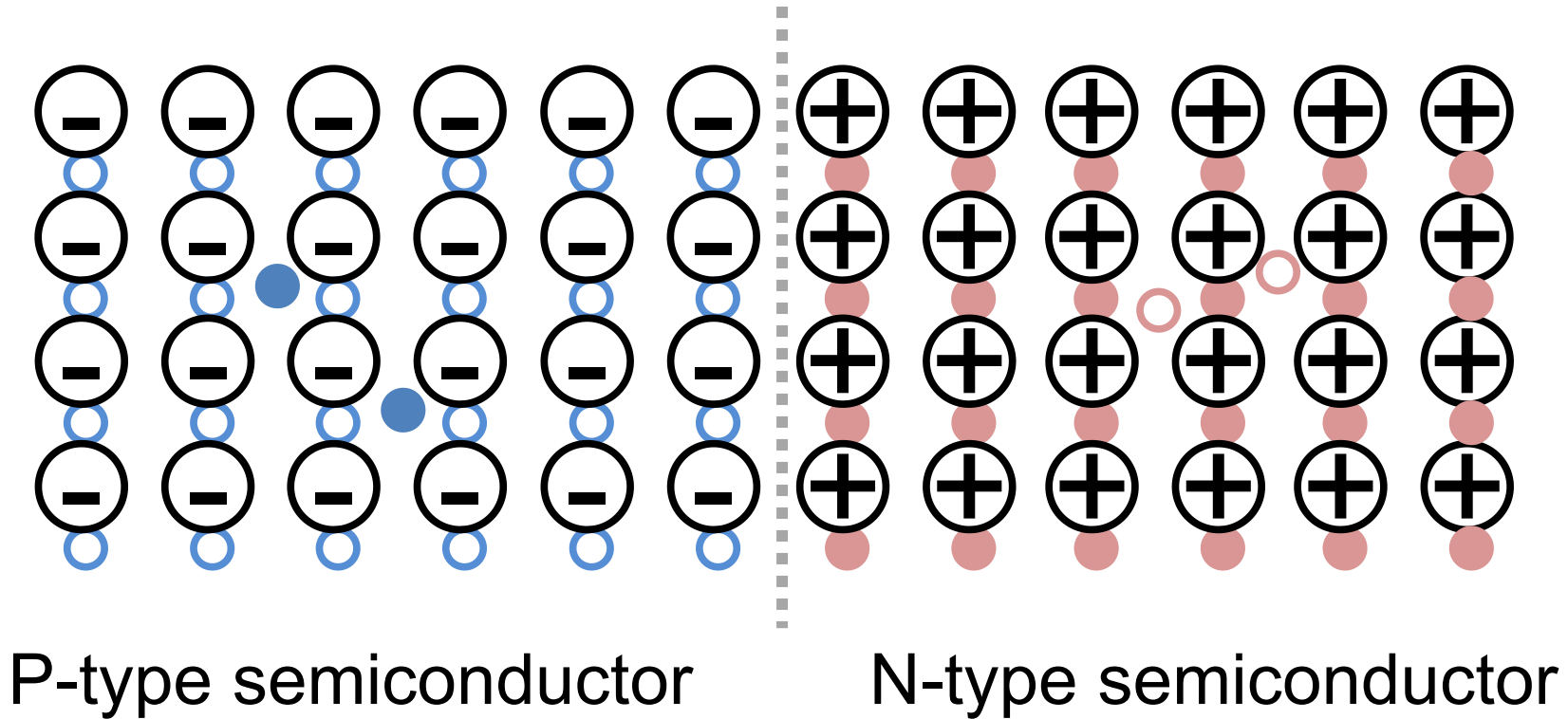
Electron acceptor: 硼(B), 3 electrons in outmost shell

Holes are majority carriers, electrons are minority carriers
空穴是多数载流子, 电子是少数载流子

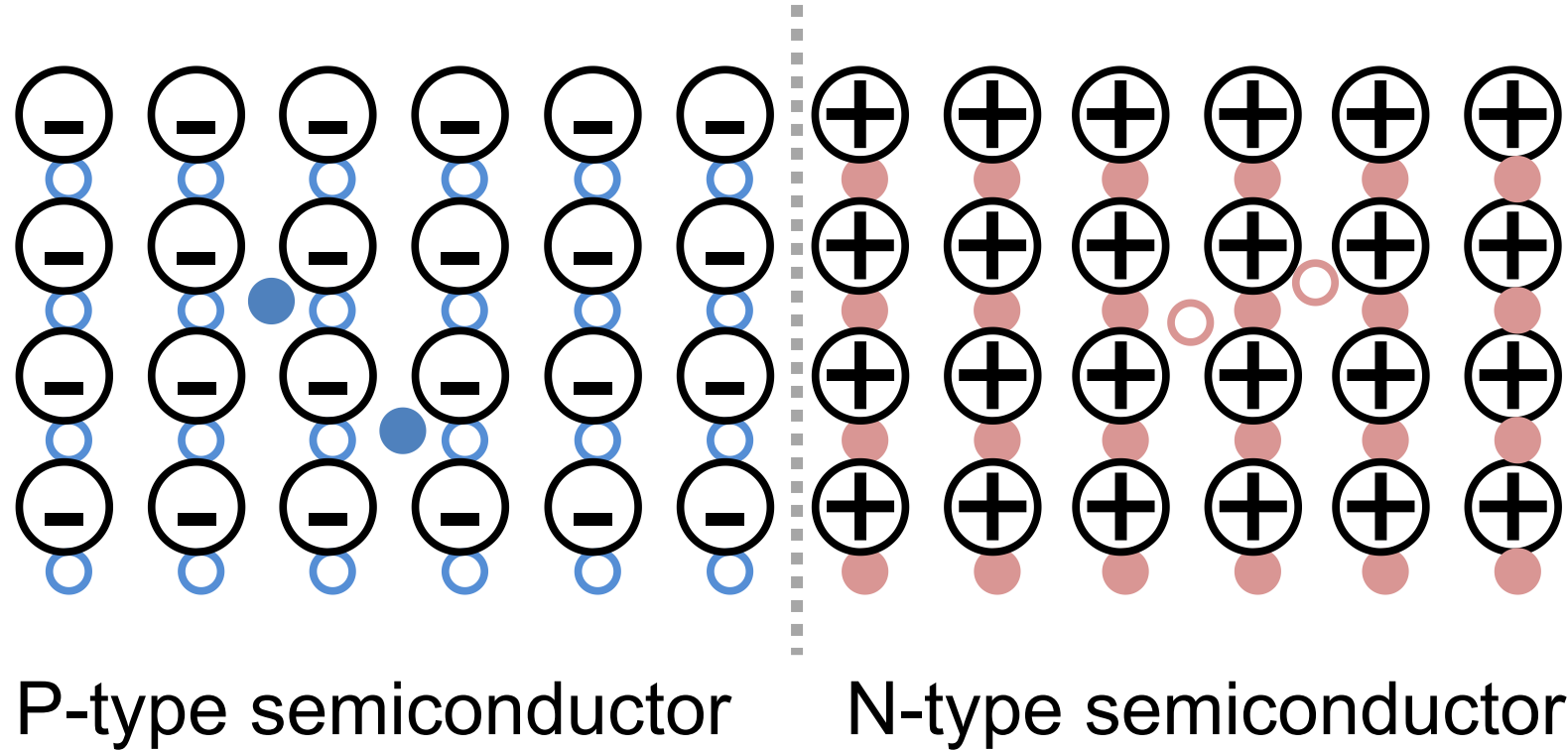
1.1.3 PN junction PN结



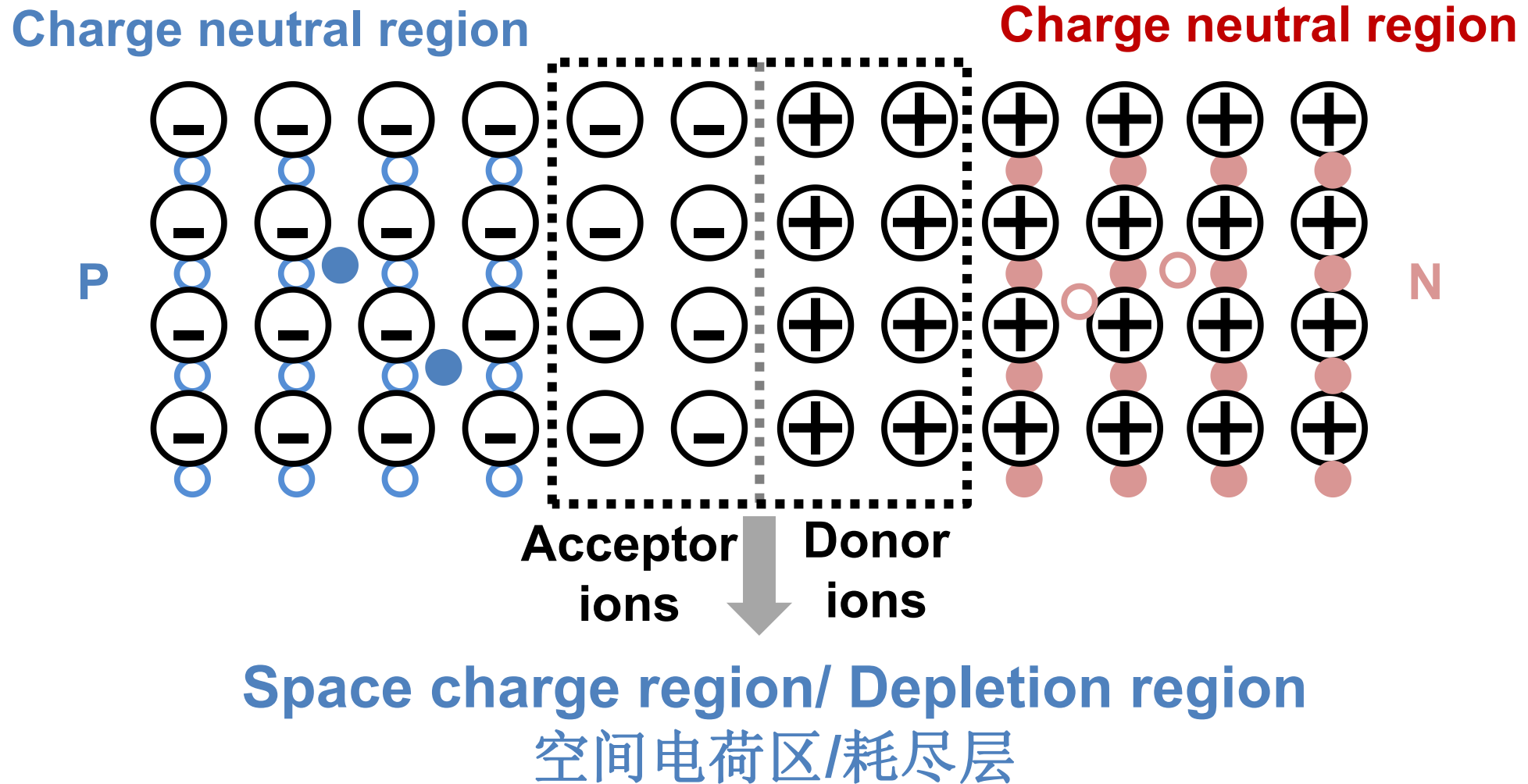
Electrons diffusive from N-type region to P-type region

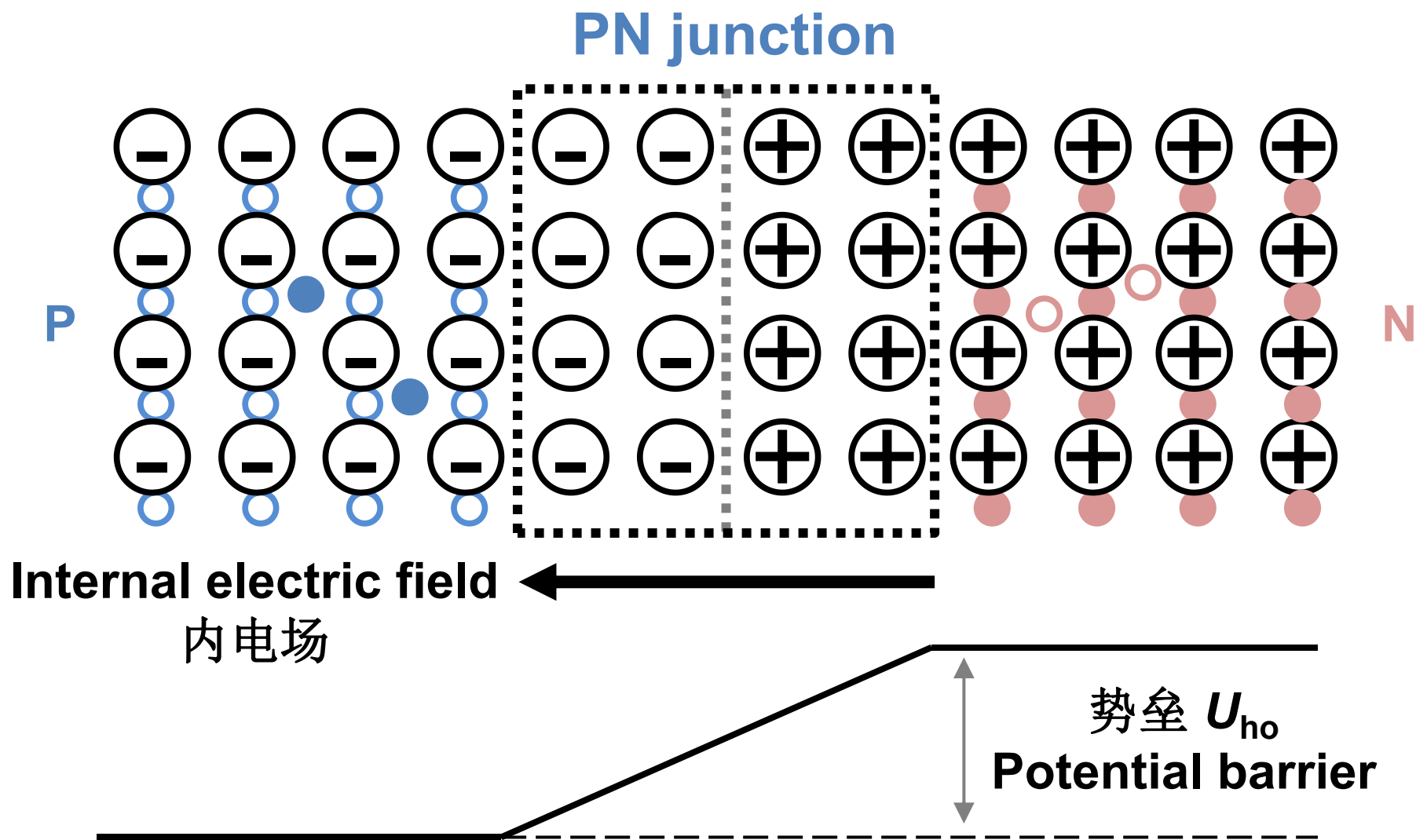


Holes diffusive from P-type region to N-type region



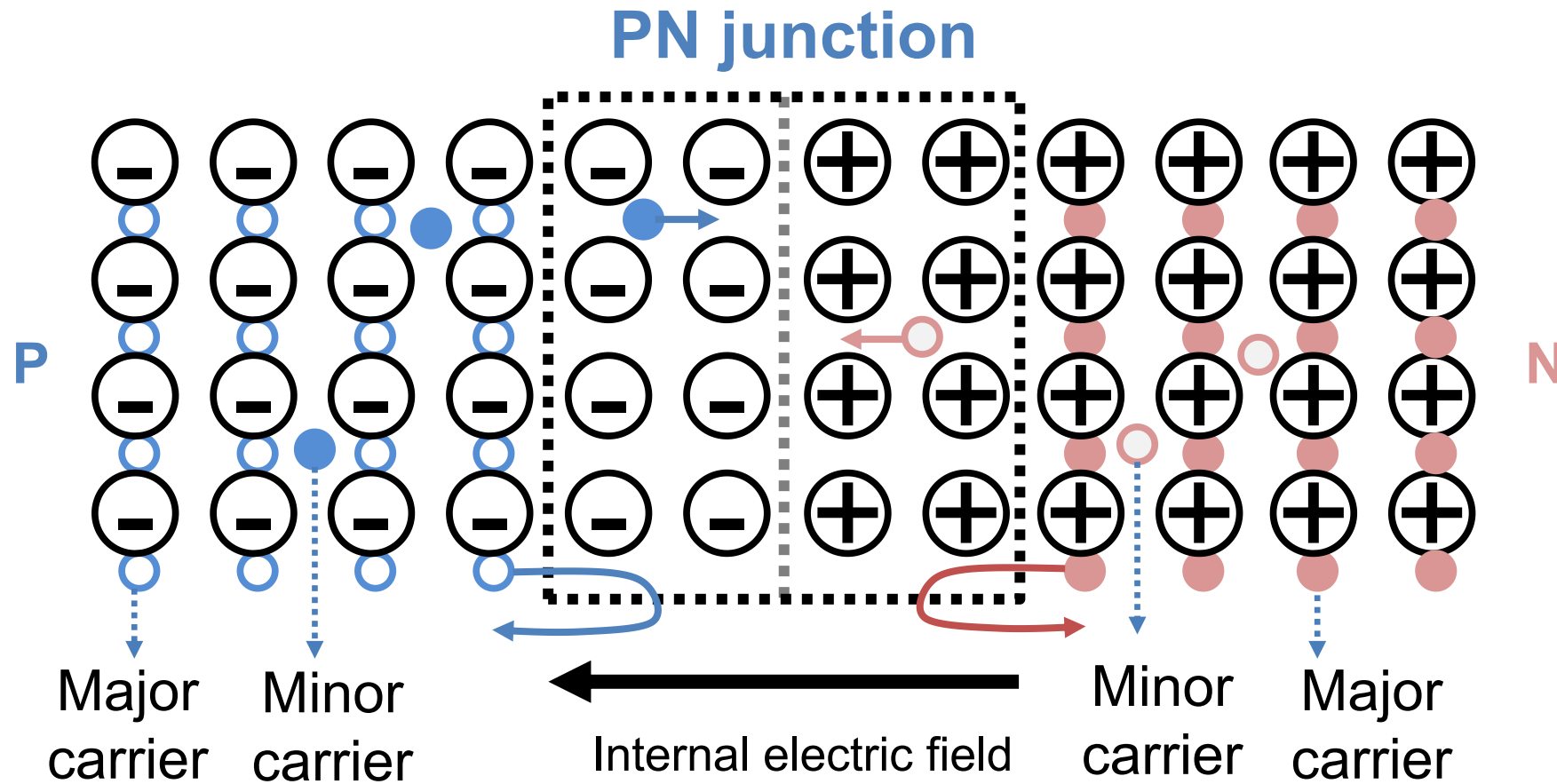
Electrons and holes recombine near the PN interface. When the system is equilibrium, only positive and negative ions left near the interface.





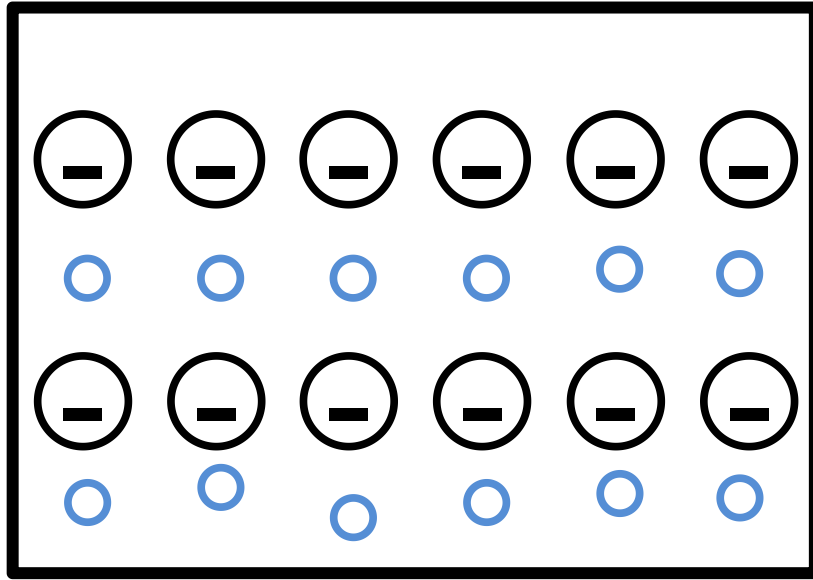
Block the diffusion of major carriers, accelerate the drift of minor carriers. 阻碍多数载流子的扩散，加速少数载流子的漂移。

**Block the diffusion of major carriers.
Accelerate the drift of minor carriers.**



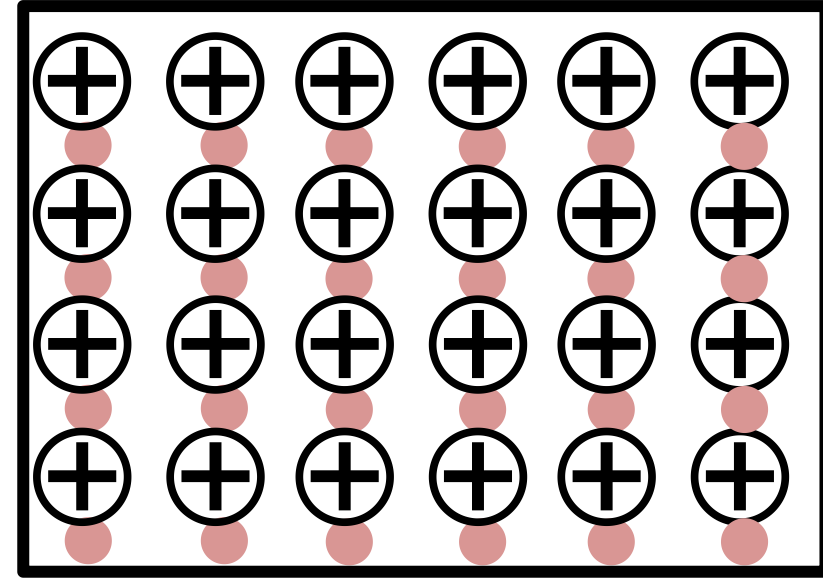
Asymmetric PN junction

Q: If they are in contact, whose space charge region is thicker?



Slightly doped P-type

Ion and free hole
concentration is low.

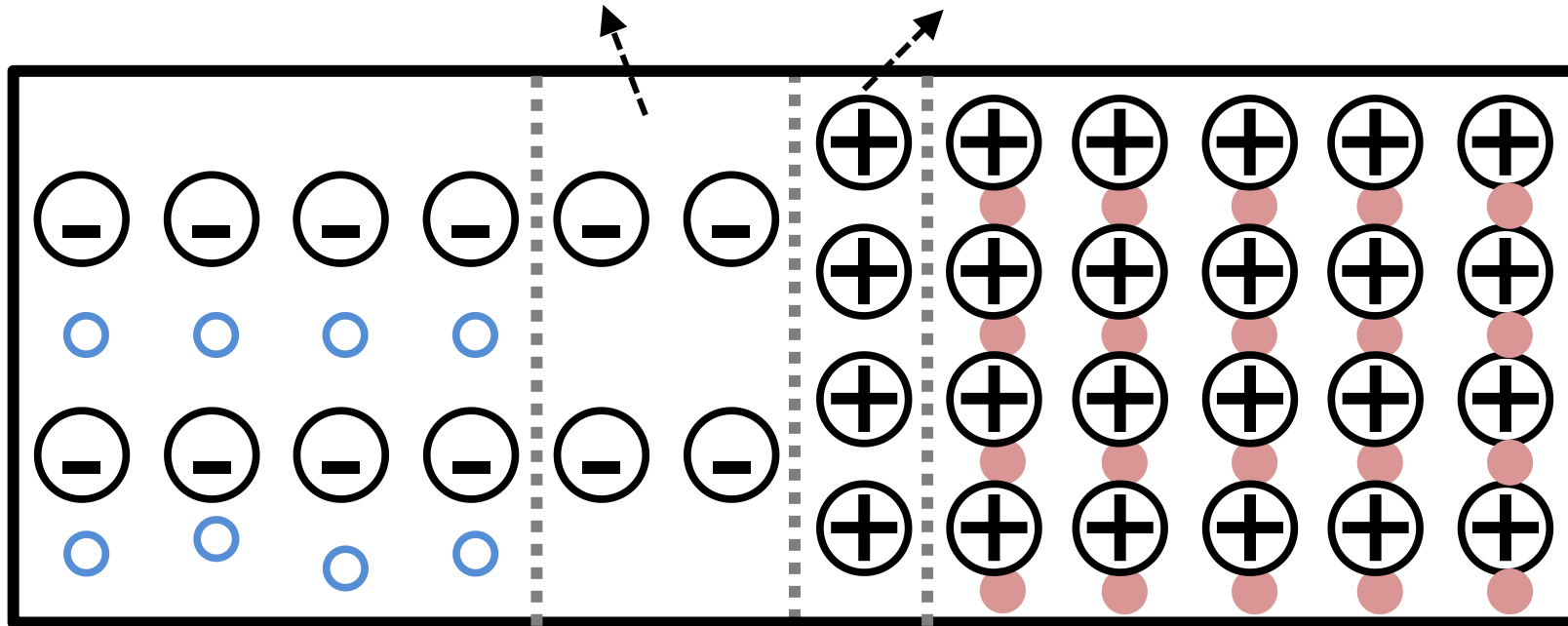


Heavily doped N-type

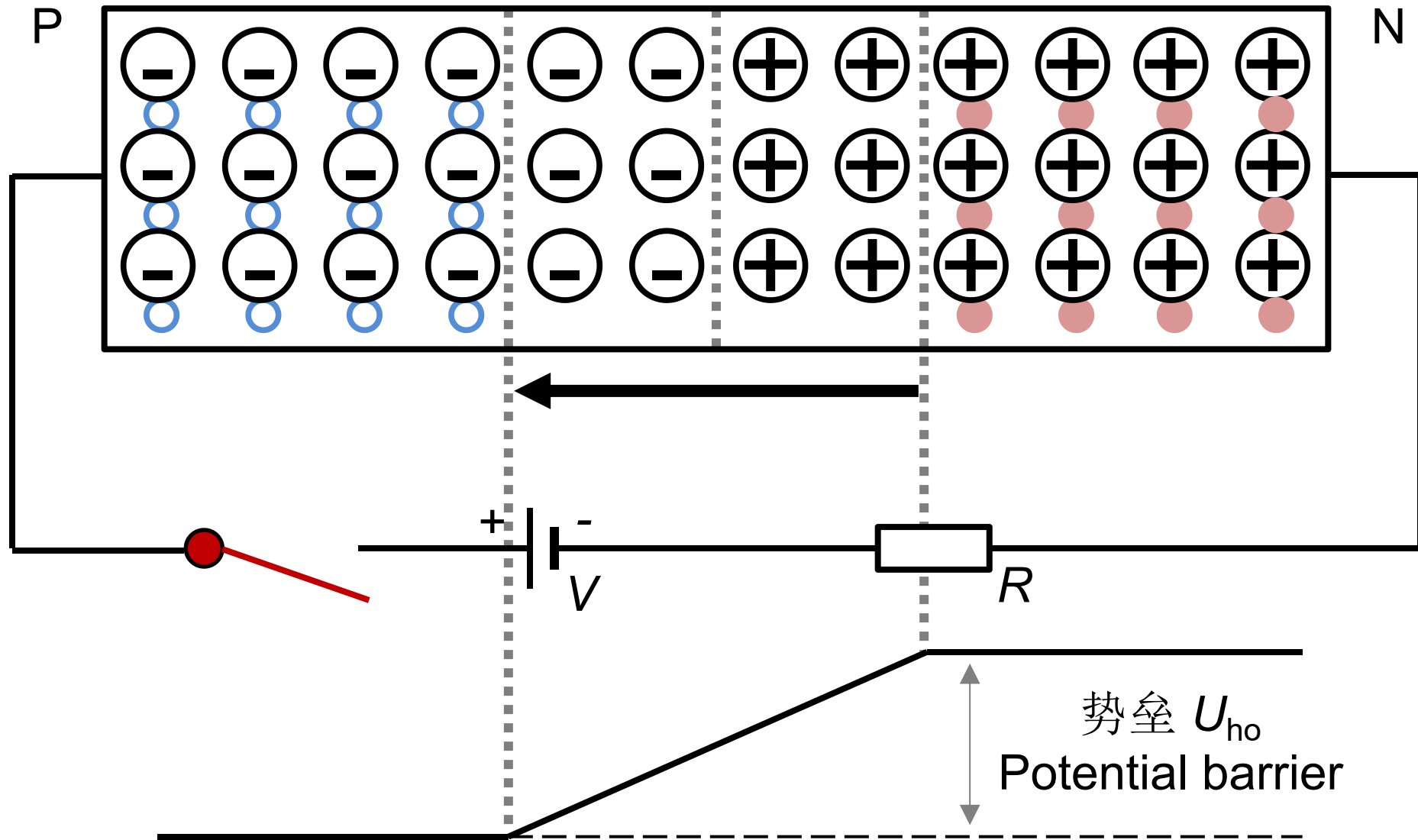
Ion and free electron
concentration is high.

Space charge region is thick

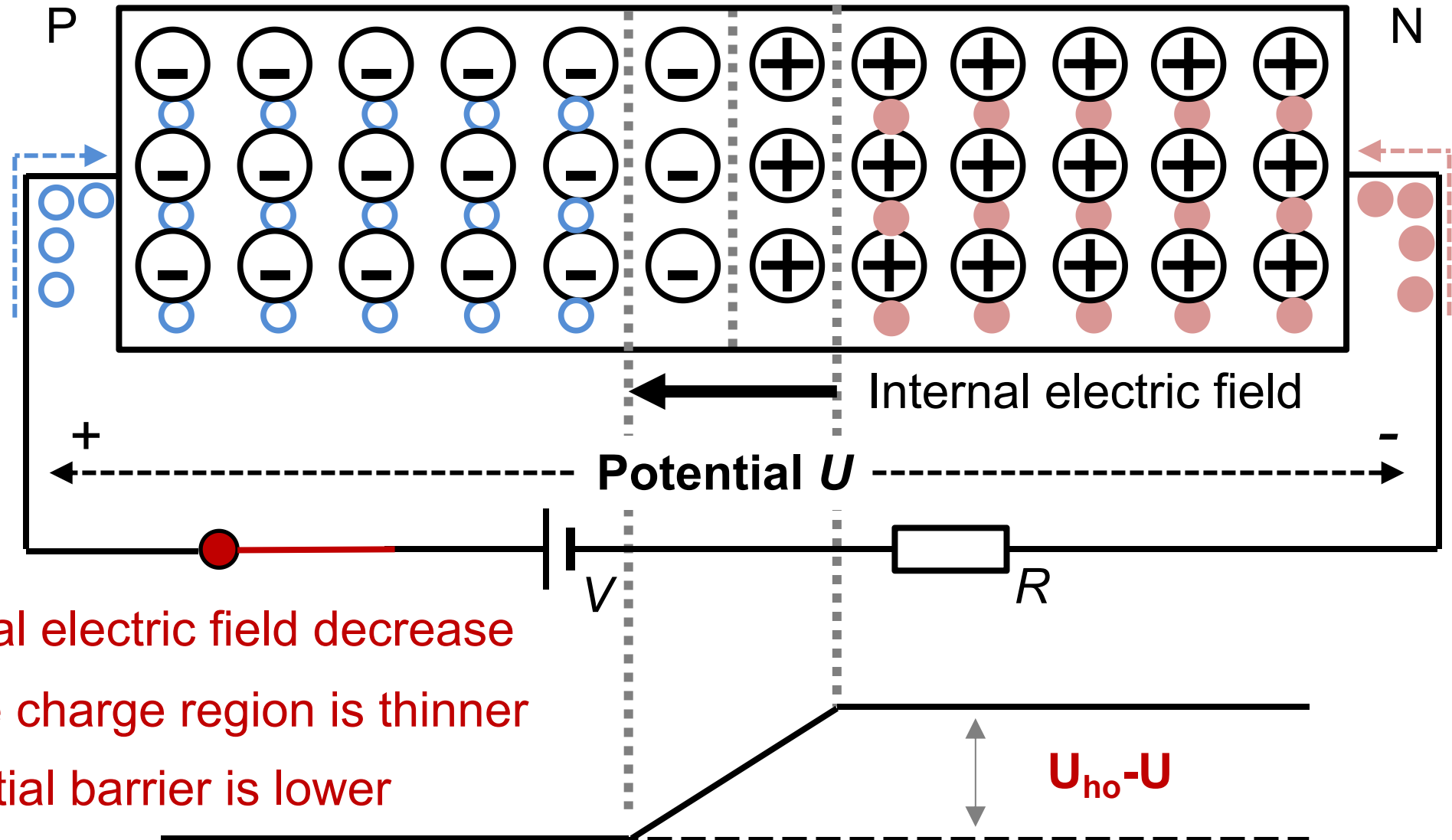
Space charge region is thin



The electrical properties of PN junction



Forward bias 正向偏置电压



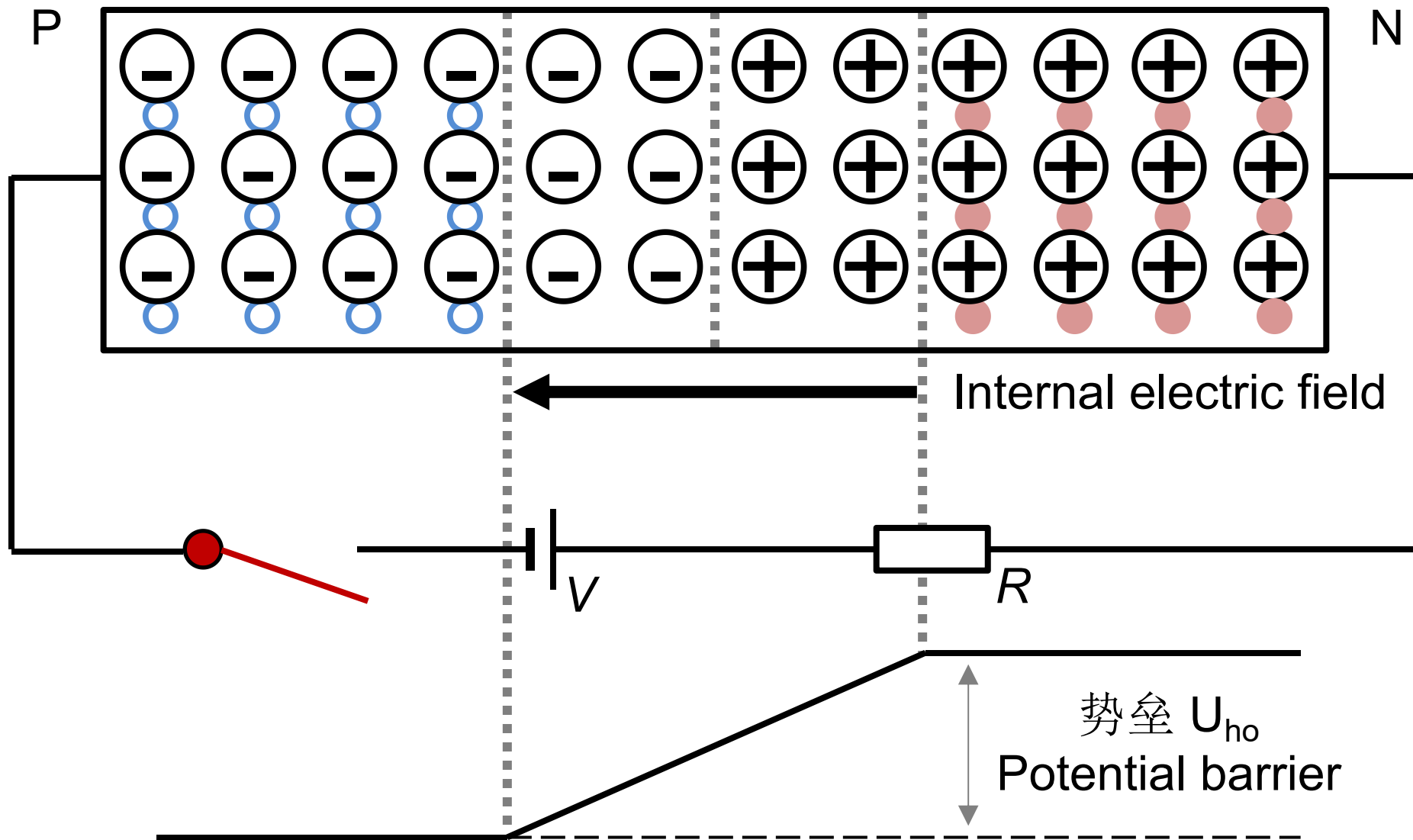
Internal electric field decrease

Space charge region is thinner

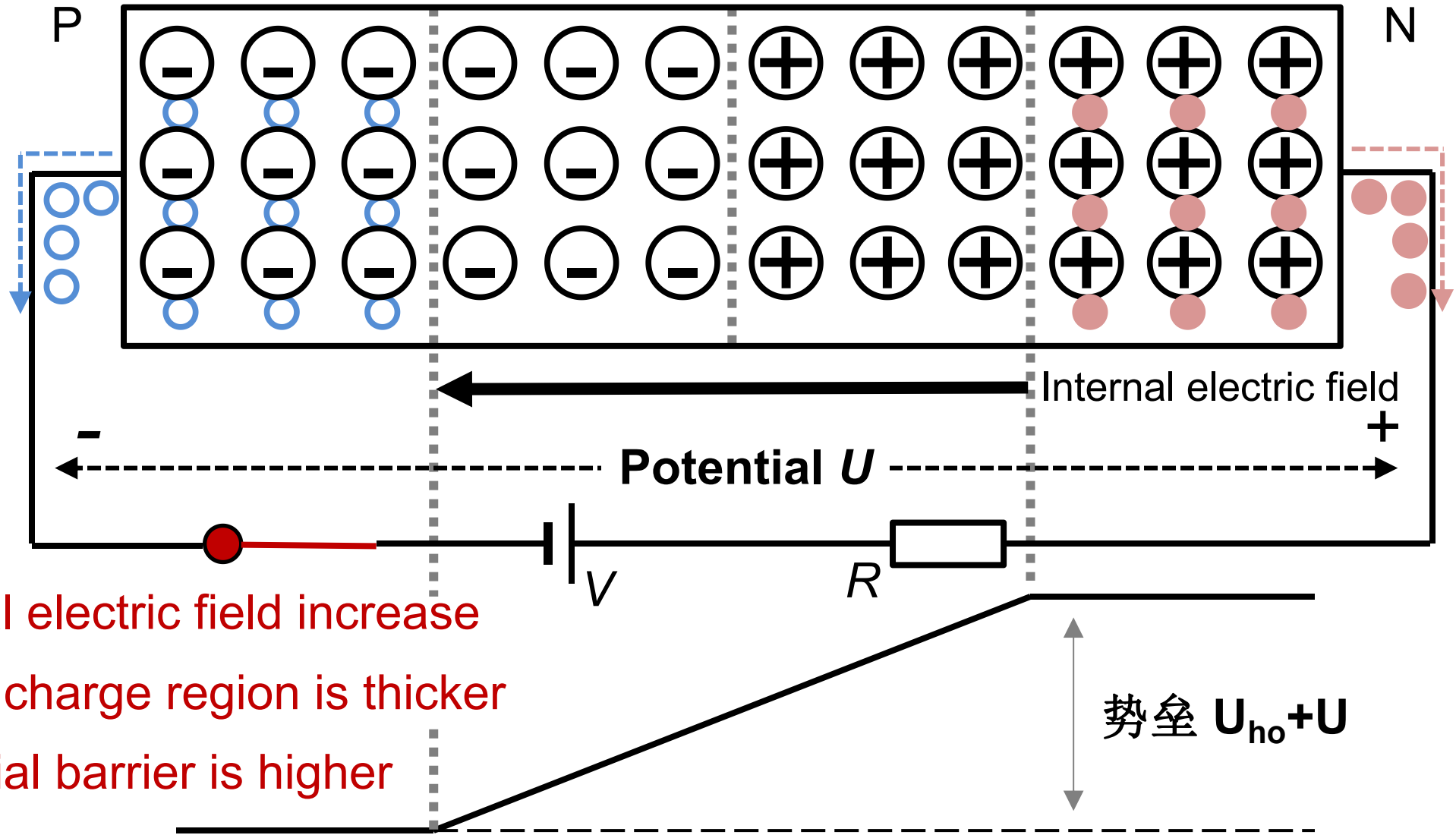
Potential barrier is lower

PN-junction is on with low resistance

Reverse bias 反向偏置电压



Reverse bias 反向偏置电压



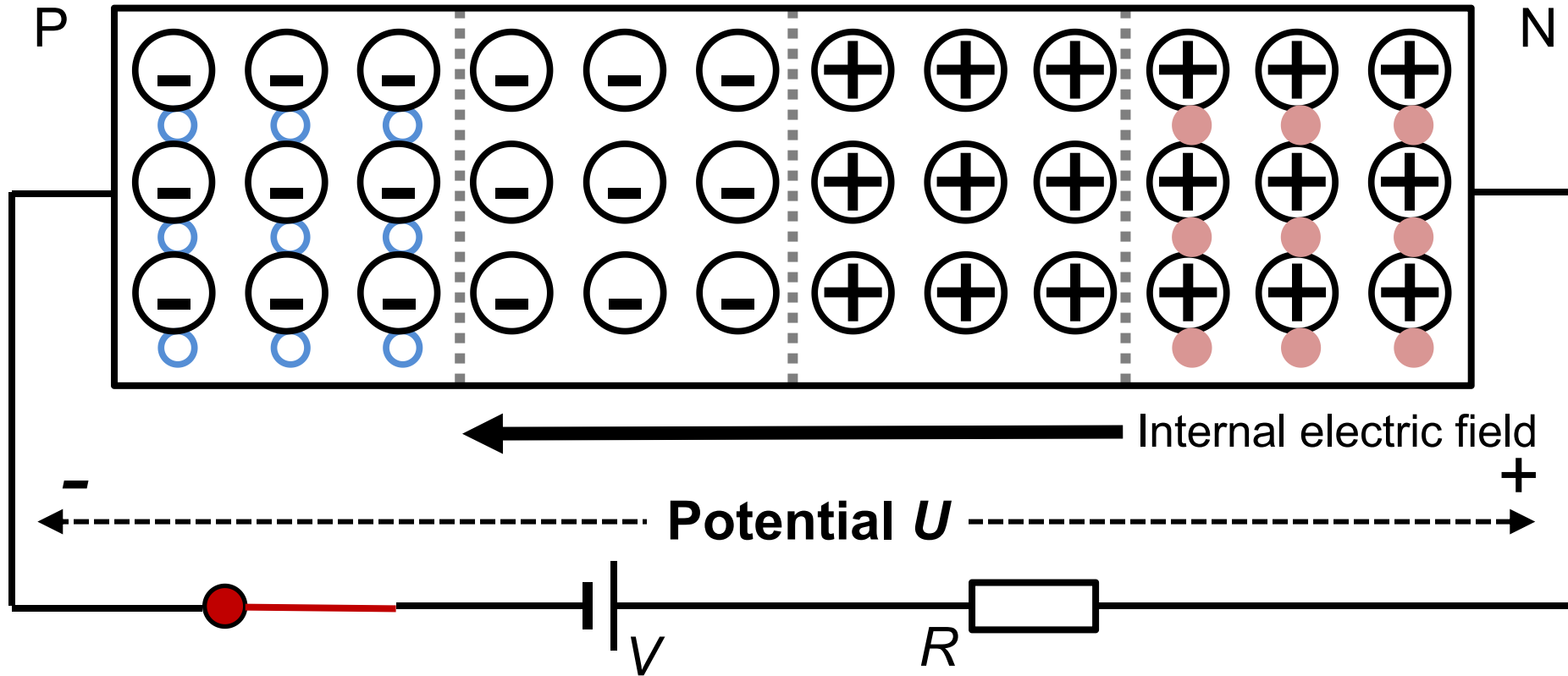
Internal electric field increase

Space charge region is thicker

Potential barrier is higher

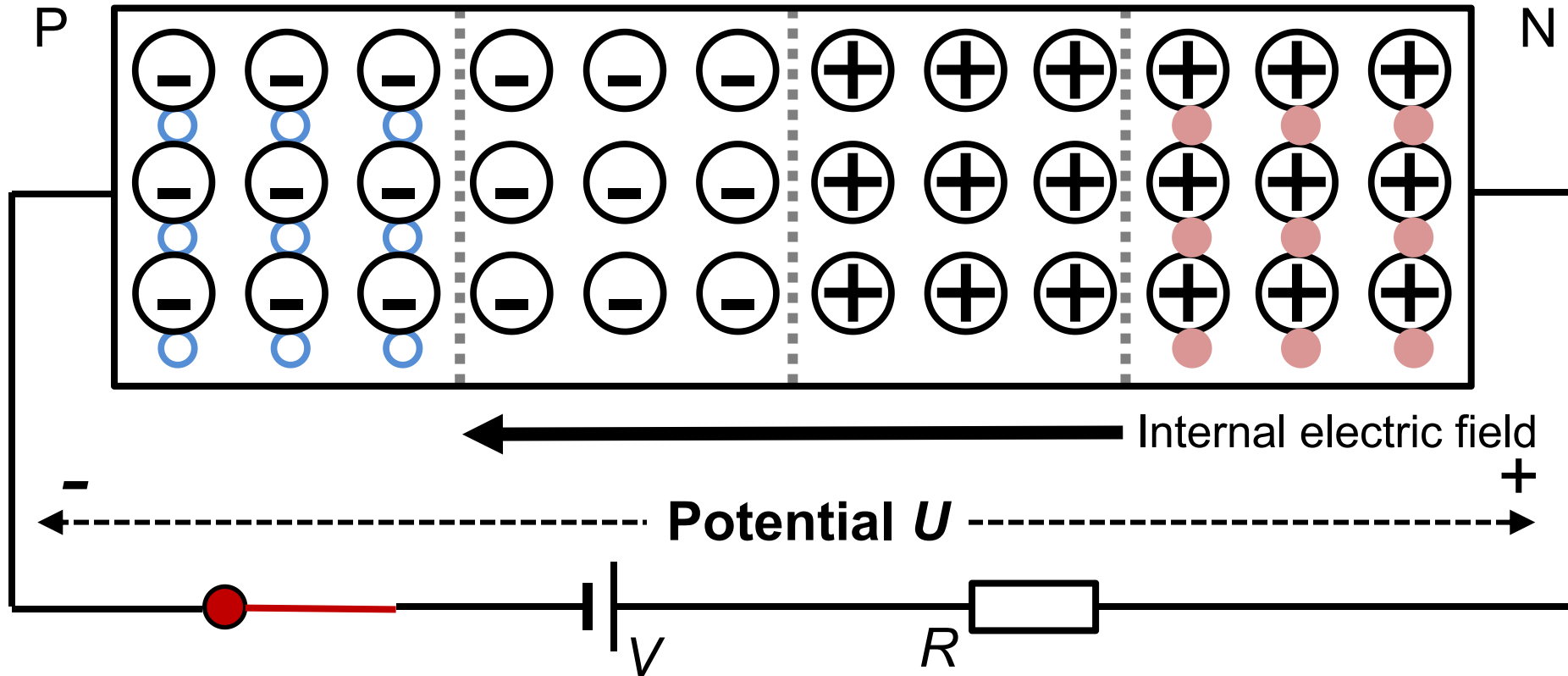
PN-junction is off with high resistance

Reverse bias 反向偏置电压



Q: Will there still be current in reverse bias?

Reverse bias 反向偏置电压



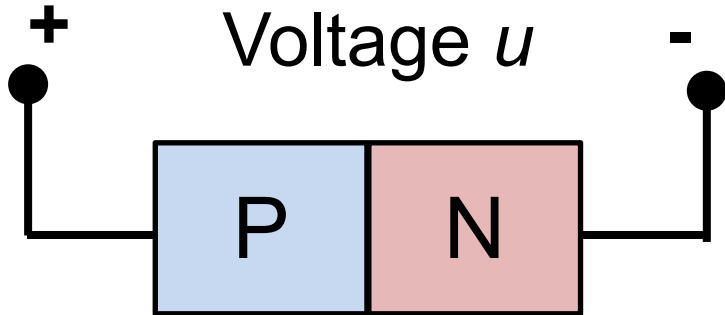
The diffusion of major carriers are further blocked.

The drift of minor carriers are further enhanced.

There will be a small current, which is called the **reverse saturation current**.

反向饱和电流

Current equation of PN junction



$$i = I_S (e^{\frac{u}{U_T}} - 1)$$

I_S : Reverse saturation current

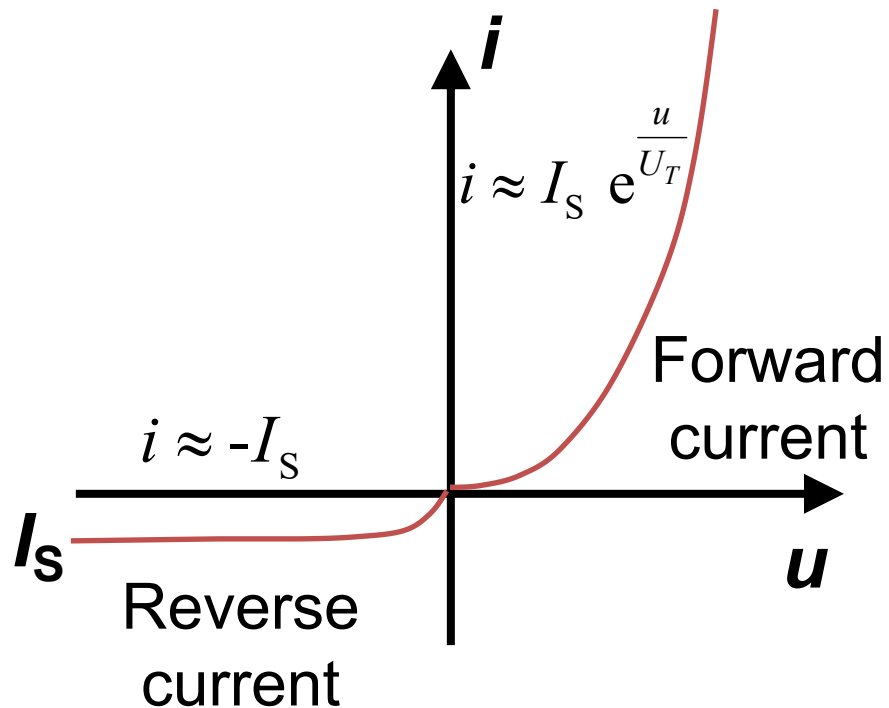
U_T : Thermal voltage $U_T = kT/q$

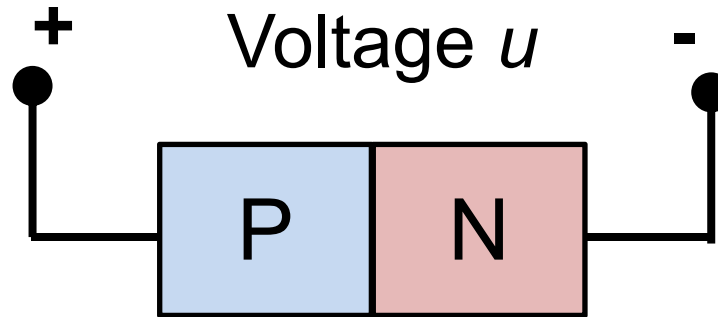
k : Boltzmann constant

q : an electron charge

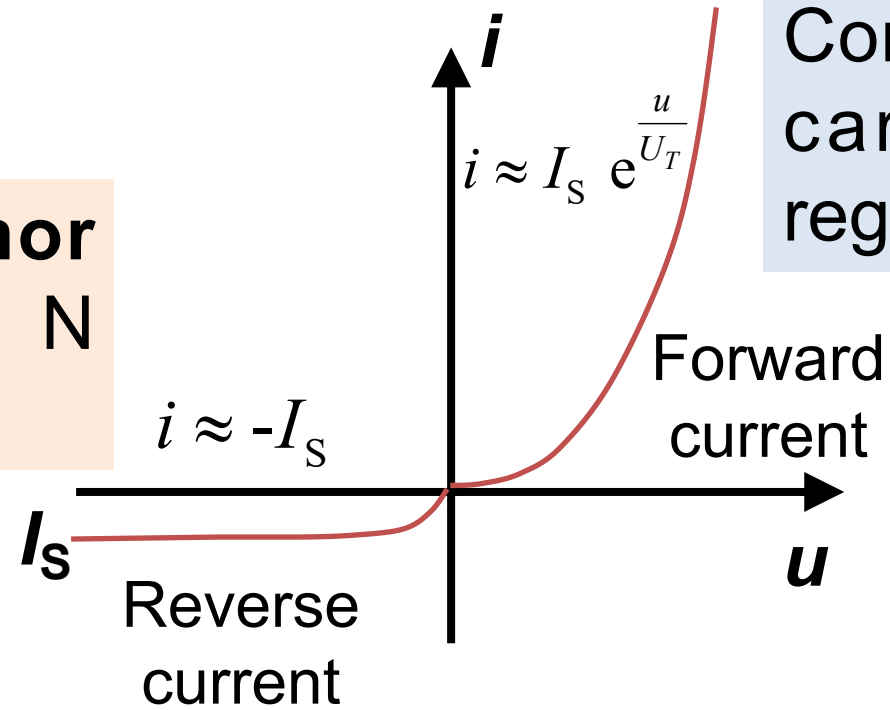
T : temperature (Kelvin)

**At room temperature ($T=300\text{K}$),
 $U_T=26\text{ mV}$**





Contributed by **minor** carriers in P and N region



Contributed by **major** carriers in P and N region